

SS1 FURTHER MATHEMATICS LESSON NOTES

FIRST TERM

WEEK 1: SETS (I)

TOPIC: INTRODUCTION TO SETS

INTRODUCTION

In mathematics, **Sets** are the *foundation* of all topics — from algebra and geometry to logic and probability.

Think of a **set** as a *container* or *collection* that holds things (called **elements**) which share a common property.

Illustration 1 — Concept of a Set

Set of Fruits (F)

$F = \{\text{🍏}, \text{🍌}, \text{🍇}, \text{🍉}\}$

Each fruit is a distinct element of the set F.

Sets are used everywhere:

- The set of students in a class.
 - The set of natural numbers.
 - The set of letters in the English alphabet.
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1. DEFINITION OF A SET

A **set** is a *well-defined collection of distinct objects* (called elements).

Keywords:

- **Well-defined:** You can clearly say if something belongs or doesn't belong to the set.
- **Distinct:** No repetition of elements.

✓ **Example (Well-defined):** $A = \{\text{The months of the year}\} \rightarrow \{\text{January, February, ..., December}\}$

✗ **Not Well-defined:** $B = \{\text{Beautiful flowers}\} \rightarrow \text{The word } \textit{beautiful} \text{ is subjective.}$

2. SET NOTATION

There are two main ways to describe sets:

A. Roster (Listing) Method

Elements are listed one by one, enclosed in curly brackets $\{ \}$.

Example: $A = \{2, 4, 6, 8, 10\}$

B. Set-builder Notation

We describe the property that defines the elements.

Example: $A = \{x \mid x \text{ is an even number, } 0 < x \leq 10\}$ (Read as: *A is the set of all x such that x is an even number between 0 and 10.*)

COMMON SYMBOLS IN SET THEORY

Symbol	Meaning	Example
$\in$	“is an element of”	$4 \in \{1,2,3,4,5\}$
$\notin$	“is not an element of”	$7 \notin \{1,2,3,4,5\}$
$\subset$	Subset	$A \subset B$
$\subseteq$	Subset (may be equal)	$A \subseteq A$
$\emptyset$	Empty Set	$\emptyset \subset A$
U	Universal Set	All elements considered

3. TYPES OF SETS

Type	Description	Example
Null Set ($\emptyset$)	Contains no elements	$\emptyset = \{ \}$ (humans who can fly unaided)
Singleton Set	Has only one element	$\{\text{The President of Nigeria}\}$
Finite Set	Countable number of elements	$\{1, 2, 3, 4, 5\}$
Infinite Set	Endless elements	$\{1, 2, 3, 4, \dots\}$
Equal Sets	Contain exactly the same elements	$\{1,2,3\} = \{3,1,2\}$
Equivalent Sets	Contain same number of elements	$\{a,b,c\}$ and $\{1,2,3\}$

Illustration 2 — Types of Sets

- [$\emptyset$] $\rightarrow$ Null Set
- [$\{A\}$] $\rightarrow$ Singleton Set
- [$\{1,2,3\}$] $\rightarrow$ Finite Set
- [$\{1,2,3,4,\dots\}$] $\rightarrow$ Infinite Set

4. UNIVERSAL SET (U)

The **Universal Set (U)** is the set of all objects under discussion.

Example: If $U = \{1,2,3,4,5,6,7,8,9,10\}$, and $A = \{2,4,6,8,10\}$, then A is a subset of U.

Note: The Universal Set depends on context. For geometry, U may be “all triangles”; for probability, “all possible outcomes”.

5. SUBSETS ($\subset$, $\subseteq$)

If every element of A is also in B, then A is a subset of B.

Example: $A = \{1,2\}$ $B = \{1,2,3,4\} \Rightarrow A \subset B$

Every set has:

- **Proper subsets:** Excludes the full set itself.
 - **Improper subsets:** Includes the set itself and $\emptyset$.
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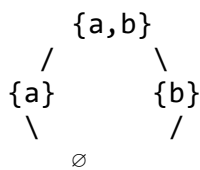
6. POWER SET (P(A))

The **power set** of a set A is the *set of all possible subsets of A*.

If $n(A)$ = number of elements in A, then **$n(P(A)) = 2^n$** .

Example: $A = \{a, b\}$ Subsets = $\emptyset$, $\{a\}$, $\{b\}$, $\{a,b\}$ Hence, $P(A) = \{\emptyset, \{a\}, \{b\}, \{a,b\}\}$ and $n(P(A)) = 4$.

Illustration 3 — Power Set Tree



WORKED EXAMPLES

Example 1: Let $V = \{a, e, i, o, u\}$ a) Write in set-builder form. b) Is it finite or infinite? c) Find $n(V)$. d) Is $a \in V$? e) Is $b \in V$?

✓ **Solution:** a) $V = \{x \mid x \text{ is a vowel in English alphabet}\}$ b) Finite (since we can count all 5 vowels) c) $n(V) = 5$ d) Yes, $a \in V$ e) No, $b \notin V$

Example 2: Let $C = \{\text{red, green}\}$ a) List all subsets. b) Write the power set $P(C)$. c) Find $n(P(C))$.

✓ **Solution:** Subsets $= \emptyset, \{\text{red}\}, \{\text{green}\}, \{\text{red, green}\}$ $P(C) = \{\emptyset, \{\text{red}\}, \{\text{green}\}, \{\text{red, green}\}\}$
 $n(P(C)) = 4$

REVISION QUESTIONS

1. Which of the following are well-defined? a) The set of interesting subjects in SS1. b) The set of rivers in Nigeria. c) The set of tasty meals.
 2. List all elements of $P = \{x \mid x \text{ is a prime number and } 10 < x < 25\}$.
 3. Let $A = \{1, \{2,3\}, 4\}$. Is $\{2,3\} \in A$? Is $2 \in A$?
 4. If $n(X) = 5$, find the number of subsets and proper subsets of X .
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ASSIGNMENT (TAKE-HOME)

1. Classify as finite or infinite: a) All leaves on trees in Lagos. b) $A = \{x \mid x \text{ is an integer } < 0\}$. c) $B = \{\text{people who are 300 years old}\}$.
 2. Let $U = \{x \mid 0 < x \leq 12, x \in \mathbb{N}\}$, $A = \{\text{factors of 12}\}$. a) List elements of U . b) List elements of A . c) Is $A \subset U$?
 3. If $n(P) = k$, find $n(P(P))$.
 4. A set X has 63 proper subsets. Find the number of elements in X .
 5. Given $S = \{a, b, c, d, e\}$, list all subsets with exactly 3 elements.
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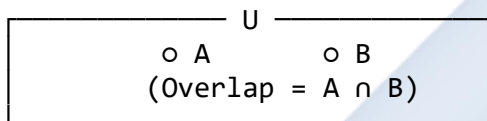
WEEK 2: SETS (II)

TOPIC: SET OPERATIONS AND VENN DIAGRAMS

INTRODUCTION

Last week, we described single sets. Now, we'll learn how **sets interact** — combining, overlapping, and excluding elements. The visual tool for this is the **Venn Diagram**, invented by **John Venn (1880)**.

Illustration 4 — Basic 2-Set Venn Diagram



1. SET OPERATIONS

Operation	Symbol	Definition	Example
Union	$\cup$	Elements in A or B or both	$A = \{1,2,3\}, B = \{3,4,5\}, A \cup B = \{1,2,3,4,5\}$
Intersection	$\cap$	Elements common to both	$A \cap B = \{3\}$
Complement	A'	Elements in U but not in A	If $U = \{1-5\}, A = \{1,2,3\}, A' = \{4,5\}$
Difference	$A \setminus B$	Elements in A but not in B	$A \setminus B = \{1,2\}$

Illustration 5 — Union and Intersection

Union ($A \cup B$): Both circles shaded

Intersection ($A \cap B$): Only the overlap shaded

2. PRINCIPLE OF INCLUSION-EXCLUSION

For two sets:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

For three sets:

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

3. VENN DIAGRAMS

A **Venn Diagram** shows sets pictorially.

- The rectangle = Universal Set (U).
- Circles = Subsets (A, B, C).
- Overlaps = Intersections.

Illustration 6 — 3-Set Venn Diagram Layout

$A \cap B \cap C$ (center)
 $A \cap B$ only
A only
B only
C only
Outside region = $(A \cup B \cup C)'$

4. WORKED EXAMPLES

Example 1 — 2-Set Problem

In a class of 35 students, 20 offer Physics (P) and 17 offer History (H). 5 offer neither.

Find: a) Students who offer both. b) Students who offer History only.

✓ **Solution:** $n(U)=35$, $n(P)=20$, $n(H)=17$, “Neither”=5

$n(P \cup H)=35-5=30$ Use inclusion-exclusion: $n(P \cup H)=n(P)+n(H)-n(P \cap H)$ $30=20+17-n(P \cap H)$
 $\Rightarrow n(P \cap H)=7$

a) **Both = 7** b) **History only = $n(H)-n(P \cap H)=17-7=10$**

Illustration 7 — 2-Set Diagram

$(A \cap B=7)$
A=20 total $\rightarrow$ A only=13
B=17 total $\rightarrow$ B only=10
Outside=5

Example 2 — 3-Set Problem

Data	Value
$n(U)$	100
$n(F)$	40
$n(C)$	30
$n(P)$	35

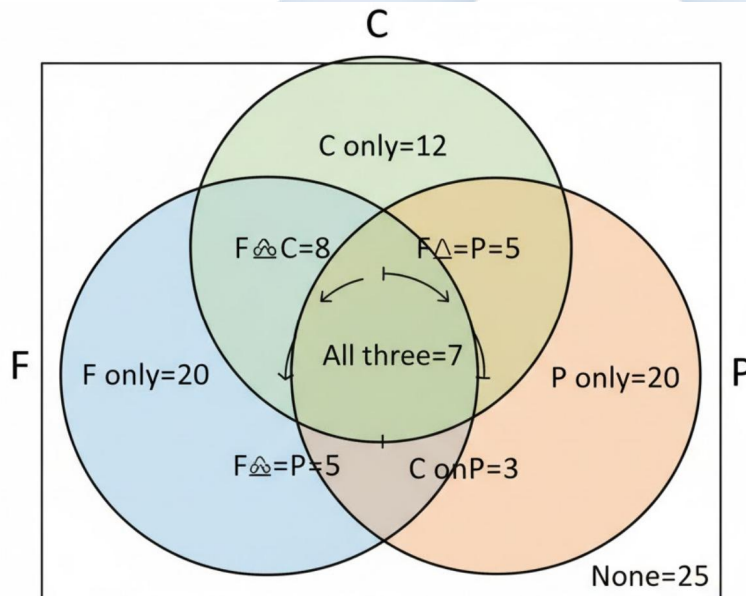
Data	Value
$n(F \cap C)$	15
$n(F \cap P)$	12
$n(C \cap P)$	10
$n(F \cap C \cap P)$	7

✓ **Solution:**

- $n(F \cap C \text{ only}) = 15 - 7 = 8$
- $n(F \cap P \text{ only}) = 12 - 7 = 5$
- $n(C \cap P \text{ only}) = 10 - 7 = 3$
- $n(F \text{ only}) = 40 - (8 + 5 + 7) = 20$
- $n(C \text{ only}) = 30 - (8 + 3 + 7) = 12$
- $n(P \text{ only}) = 35 - (5 + 3 + 7) = 20$

Total inside = 75 “None” = $100 - 75 = 25$

Illustration 8 — 3-Set Venn Diagram



5. DE MORGAN'S LAWS

For any sets A and B:

1. $(A \cup B)' = A' \cap B'$
2. $(A \cap B)' = A' \cup B'$

Illustration: Overlapping circles showing how complements flip union/intersection.

REVISION QUESTIONS

1. If $U=\{1,2,\dots,10\}$, $A=\{1,3,5,7,9\}$, $B=\{2,3,5,7\}$, find: a) $A \cup B$ b) $A \cap B$ c) A' d) $B \setminus A$
2. Shade region $B \cap A'$ in a Venn diagram.
3. State De Morgan's Laws.
4. If 15 play Football, 12 play Basketball, and 5 both, how many students are there if all play at least one?

ASSIGNMENT (TAKE-HOME)

1. Given $U=\{a,b,c,d,e,f,g,h\}$, $X=\{a,b,d,e\}$, $Y=\{b,e,f,g\}$: Find: a) $X' \cap Y$ b) $(X \cup Y)'$ c) Verify $(X \cup Y)' = X' \cap Y'$
2. In a group of 50 people, 30 speak English, 25 speak French, and 10 speak neither: a) Draw a Venn diagram. b) Find number who speak both. c) Find English only.
3. A survey of 60 fans: 28 watch Football (F), 22 Basketball (B), 25 Cricket (C), 10 $F \cap B$, 8 $F \cap C$, 7 $B \cap C$, 3 all three. a) Fill all 8 regions. b) How many watch exactly two? c) How many watch none?
4. If $n(A \setminus B)=15$, $n(B \setminus A)=10$, $n(A \cap B)=5$, find $n(A \cup B)$.
5. Draw and shade region $A \cap (B \cup C)'$.

WEEK 3: INDICES AND LOGARITHMS

TOPIC: INDICES AND LOGARITHMS

INTRODUCTION

Indices and logarithms are two powerful tools that help us express very large or very small numbers and solve exponential equations efficiently.

- **Indices (or powers)** express repeated multiplication compactly. [Example: $(2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32)$]
- **Logarithms** do the reverse of indices. If $10^2 = 100$, then $\log_{10} 100 = 2$. [This means "10 must be raised to the power of 2 to get 100."]

Understanding both is crucial for later work in **exponential growth, compound interest, and calculus**.

SECTION A: INDICES

1. LAWS OF INDICES

For any positive base a and integers m, n :

Law	Algebraic Form	Example
Multiplication	$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^7 = 128$
Division	$a^m \div a^n = a^{m-n}$	$5^6 \div 5^2 = 5^4 = 625$
Power	$(a^m)^n = a^{mn}$	$(3^2)^4 = 3^8 = 6561$
Zero Index	$a^0 = 1$	$7^0 = 1$
Negative Index	$a^{-n} = 1/a^n$	$2^{-3} = 1/8$
Fractional Index	$a^{1/n} = \sqrt[n]{a}$	$8^{1/3} = \sqrt[3]{8} = 2$
Mixed Fractional	$a^{m/n} = (\sqrt[n]{a})^m$	$27^{2/3} = (\sqrt[3]{27})^2 = 9$

ILLUSTRATION 1 – The Meaning of Indices

$$a^5 = a \times a \times a \times a \times a$$

Exponent (5) → tells how many times the **base (a)** is multiplied by itself.

$$\begin{array}{ccccccc} & a & \times & a & \times & a & \times & a & \times & a \\ & \uparrow & & & & & & \uparrow & & \\ \text{base} & & & & & & & \text{power (exponent)} & & \end{array}$$

2. APPLICATIONS OF INDICES

Example 1: Simplify $(16/81)^{-3/4}$

Solution:

$$\begin{aligned} (16/81)^{-3/4} &= (81/16)^{3/4} \\ &= \left(\frac{\sqrt[4]{81}}{\sqrt[4]{16}} \right)^3 = \left(\frac{3}{2} \right)^3 = \frac{27}{8} \end{aligned}$$

Example 2: Solve $4^x - 6(2^x) + 8 = 0$

Let $y = 2^x \Rightarrow 4^x = (2^2)^x = (2^x)^2 = y^2$.

Then:

$$\begin{aligned} y^2 - 6y + 8 &= 0 \\ (y - 2)(y - 4) &= 0 \end{aligned}$$

So, $y = 2$ or $y = 4$

Substitute back $2^x = 2 \Rightarrow x = 1$;

$$2^x = 4 \Rightarrow x = 2$$

✓ **Final Answer:** $x = 1$ or 2

3. GRAPHICAL INTERPRETATION OF EXPONENTIAL GROWTH

Function	Behavior
$y = 2^x$	Increases rapidly as x increases
$y = (1/2)^x$	Decreases (decay curve)

(Imagine two curves on a Cartesian plane: one rising steeply, one decaying.)

SECTION B: LOGARITHMS

1. DEFINITION

If $a^x = N$, then $\log_a N = x$

Read as: "logarithm of N to base a equals x."

Example:

$$10^3 = 1000 \Rightarrow \log_{10} 1000 = 3$$

2. LAWS OF LOGARITHMS

Law	Algebraic Form	Example
Product	$\log_a(MN) = \log_a M + \log_a N$	$\log_{10}(5 \times 4) = \log_{10} 5 + \log_{10} 4$
Quotient	$\log_a(M/N) = \log_a M - \log_a N$	$\log_{10}(50/2) = \log_{10} 50 - \log_{10} 2$
Power	$\log_a(M^p) = p \log_a M$	$\log_{10} 8^2 = 2 \log_{10} 8$

Corollaries:

- $\log_a a = 1$
- $\log_a 1 = 0$

3. CHANGE OF BASE

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Usually, $c = 10$ or e (for natural logs).

Corollary: $\log_a b = \frac{1}{\log_b a}$

ILLUSTRATION 2 – Relationship Between Indices and Logarithms

If $a^x = N$, then $\log_a N = x$.

$$a^x = N \quad \leftrightarrow \quad \log_a N = x$$

Example:

$$2^5 = 32 \leftrightarrow \log_2 32 = 5$$

WORKED EXAMPLE (Logarithm Equation)

Example 3: Solve for x if $\log_2 x + \log_2(x - 7) = 3$

Solution:

$$\log_2[x(x - 7)] = 3$$

$$x^2 - 7x = 8$$

$$x^2 - 7x - 8 = 0$$

$$(x - 8)(x + 1) = 0$$

Valid value: $x = 8$ (since $x > 7$) ✓

WORKED EXAMPLE (Change of Base)

Given $\log_3 2 = 0.631$, find $\log_2 9$.

$$\log_2 9 = \frac{\log_3 9}{\log_3 2}$$

$$\log_3 9 = 2 \Rightarrow \frac{2}{0.631} = 3.17$$

✓ Answer: $\log_2 9 = 3.17$

EVALUATION QUESTIONS

1. Simplify $(x^{1/2}y^2)^{1/3} \div (x^{-1/3}y)^{1/2}$
2. Solve $(1/4)^x = 32$
3. Express as a single logarithm: $\frac{1}{2} \log 25 - 2 \log 3 + \log 18$
4. Given $\log_{10} 2 = 0.3010$, find $\log_{10} 32$
5. Solve $5^{2k} + 2(5^k) - 3 = 0$

ASSIGNMENT

1. Simplify $((125)^{1/3} \times (27)^{2/3}) / ((64)^{-1/3} \times (81)^{1/4})$
2. Solve $3^{2x} - 10(3^x) + 9 = 0$
3. $\log_3 p - \log_3(p - 6) = 2$. Find p .
4. $\log_4(x + 1) + \log_4(x - 1) = \log_4 8$. Find x .
5. If $\log_2 x = a$ and $\log_2 y = b$, express $\log_2((x^3y)/\sqrt{8})$ in terms of a, b .

WEEK 4: SURDS

TOPIC: SURDS

INTRODUCTION

A **surd** is an irrational number that cannot be simplified to remove the square (or cube) root. For instance, $\sqrt{2}$, $\sqrt{3}$, and $\sqrt{5}$ cannot be expressed as exact fractions or terminating decimals.

Perfect Squares	Surds
$\sqrt{4} = 2$	$\sqrt{2}$ (irrational)
$\sqrt{9} = 3$	$\sqrt{3}$ (irrational)

Thus, a **surd** remains in root form to preserve **accuracy**.

1. TYPES OF SURDS

- **Simple Surd:** $\sqrt{7}$, $\sqrt[3]{5}$
 - **Compound Surd:** $2 + \sqrt{3}$, $5 - 2\sqrt{2}$
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2. RULES FOR MANIPULATING SURDS

Operation	Rule	Example
Addition/Subtraction	Only like surds combine	$3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$
Multiplication	$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$	$\sqrt{2} \times \sqrt{3} = \sqrt{6}$
Division	$\sqrt{a} / \sqrt{b} = \sqrt{a/b}$	$\sqrt{12} / \sqrt{3} = \sqrt{4} = 2$
Simplification	$\sqrt{a^2b} = a\sqrt{b}$	$\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}$

ILLUSTRATION 3 – Simplifying Surds

$$\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}$$

$$\begin{array}{ccc} & \sqrt{36 \times 2} & \\ & \uparrow \quad \uparrow & \\ \text{Perfect} & & \text{Left-over} \\ \text{Square} & & \text{factor} \end{array}$$

3. RATIONALIZATION

Objective: Eliminate the surd from the denominator.

Case 1: Monomial Denominator

$$\frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$$

Case 2: Binomial Denominator

$$\frac{10}{3 - \sqrt{2}} \times \frac{3 + \sqrt{2}}{3 + \sqrt{2}} = \frac{30 + 10\sqrt{2}}{7}$$

Concept: Multiply by the *conjugate* to remove the surd.

WORKED EXAMPLES

Example 1: Simplify $\sqrt{50} + 3\sqrt{8} - \sqrt{18}$

$$\sqrt{50} = 5\sqrt{2}, \quad \sqrt{8} = 2\sqrt{2}, \quad \sqrt{18} = 3\sqrt{2}$$

$$5\sqrt{2} + 6\sqrt{2} - 3\sqrt{2} = 8\sqrt{2}$$

✓ Answer: $8\sqrt{2}$

Example 3: Rationalize $\frac{\sqrt{5}}{2 + \sqrt{5}}$

Multiply by the conjugate $(2 - \sqrt{5})$:

$$\frac{\sqrt{5}(2 - \sqrt{5})}{(2 + \sqrt{5})(2 - \sqrt{5})} = \frac{2\sqrt{5} - 5}{4 - 5} = \frac{2\sqrt{5} - 5}{-1} = 5 - 2\sqrt{5}$$

✓ Answer: $5 - 2\sqrt{5}$

Example 3: Rationalize $\frac{\sqrt{5}}{2+\sqrt{5}}$

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$$\frac{\sqrt{5}(2 - \sqrt{5})}{(2 + \sqrt{5})(2 - \sqrt{5})} = \frac{2\sqrt{5} - 5}{4 - 5} = \frac{2\sqrt{5} - 5}{-1} = 5 - 2\sqrt{5}$$

✓ Answer: $5 - 2\sqrt{5}$

ILLUSTRATION 4 – Rationalization Process

Multiply by conjugate

$$\frac{1}{2 + \sqrt{3}} \times \frac{(2 - \sqrt{3})}{(2 - \sqrt{3})} = \frac{(2 - \sqrt{3})}{(4 - 3)}$$

Result: $2 - \sqrt{3}$

EVALUATION QUESTIONS

1. Define a surd.
2. Simplify $\sqrt{72}$.
3. Simplify $4\sqrt{5} + \sqrt{20} - \sqrt{45}$.
4. Rationalize $\frac{1}{2\sqrt{5}}$.
5. What is the conjugate of $5 - 3\sqrt{2}$?

ASSIGNMENT

1. Simplify $\sqrt{48} + \sqrt{108} - \sqrt{27}$.
 2. Expand $(\sqrt{6} + 2)(2\sqrt{6} - 3)$.
 3. Rationalize $\frac{12}{\sqrt{11} - \sqrt{7}}$.
 4. Express $\frac{3\sqrt{2}-1}{2\sqrt{2}+3}$ in the form $p + q\sqrt{2}$.
 5. Given $x = 1 + \sqrt{2}$, find $x^2 + \frac{1}{x^2}$.
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WEEK 6: LINEAR INEQUALITIES

TOPIC: LINEAR INEQUALITIES

INTRODUCTION

In algebra, an **inequality** is like an equation, but instead of showing that two sides are *equal*, it shows that one side is *greater* or *smaller* than the other.

An **inequality** uses the following symbols:

Symbol	Meaning	Example
$>$	greater than	$5 > 3$
$<$	less than	$4 < 9$
$\geq$	greater than or equal to	$x \geq 2$ (means x is 2 or more)
$\leq$	less than or equal to	$y \leq 5$ (means y is 5 or less)

A **linear inequality** is an inequality involving variables raised to the first power (no squares, cubes, or roots).

Example:

- $(2x + 5 > 9)$
- $(3x - 4 \leq 8)$

CONTENT BREAKDOWN

1. LINEAR INEQUALITIES IN ONE VARIABLE

These are inequalities with just one unknown (variable).

Form: $[ax + b > c]$ $[ax + b \leq c]$

You solve them like normal linear equations **except** for one crucial rule:

THE GOLDEN RULE: When multiplying or dividing both sides of an inequality by a **negative number**, you must **reverse the inequality sign**.

Example of the Golden Rule

$6 > 4$ (True)

Multiply both sides by -2:

$-12 > -8$ (False!) $\Rightarrow -12 < -8$ (True!)

So: ✓ If you multiply/divide by a **negative**, reverse the inequality.

Representation on the Number Line

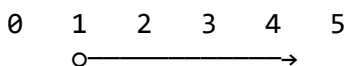
To represent inequalities:

- **Open circle** ($\circ$) $\rightarrow$ for *less than* ($<$) or *greater than* ($>$)
- **Closed circle** ($\bullet$) $\rightarrow$ for *less than or equal to* ($\leq$) or *greater than or equal to* ($\geq$)

Example:

$$[x > 2]$$

Graph:



Explanation: Open circle at 2 (because 2 is *not included*), arrow to the right showing all numbers *greater than* 2.

2. LINEAR INEQUALITIES IN TWO VARIABLES

These take the form:

$$ax + by > c \quad \text{or} \quad ax + by \leq c$$

Here, (x) and (y) are variables, and the solution is a **region on the coordinate plane**.

Steps for Graphing Linear Inequalities:

1. **Convert inequality to an equation:** Replace inequality sign with "=" to get the **boundary line**.
 2. **Draw the boundary line:**
 - Use **dashed line** for $<$ or $>$ (line not included).
 - Use **solid line** for $\leq$ or $\geq$ (line included).
 3. **Test a point** (usually the origin (0, 0)) to know which region to shade:
 - Substitute into the original inequality.
 - If true, shade the region containing (0, 0).
 - If false, shade the opposite region.
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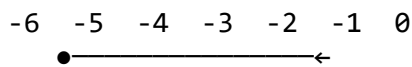
ILLUSTRATIVE DIAGRAMS

Diagram 1: Example of Linear Inequality in One Variable

Solve ($5 - 2x \geq 11$).

$$[5 - 2x \geq 11] [-2x \geq 6] \text{ Divide both sides by } -2 \text{ (reverse the sign): } [x \leq -3]$$

Graph:



Explanation: Closed circle at -3 because it includes -3 ($\leq$). Arrow to the left shows all numbers less than -3.

Diagram 2: Example of Linear Inequality in Two Variables

$$[2x - y < 3]$$

Step 1: Boundary Line Equation: $(2x - y = 3)$

Rearrange for (y) : $[y = 2x - 3]$ This is a line with:

- y-intercept = -3
- Slope = 2

Step 2: Choose points:

x	y
0	-3
2	1

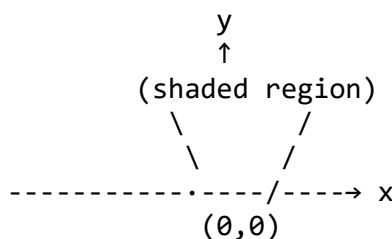
Step 3: Draw the line (dashed)

Step 4: Test the point (0,0):

$$2(0) - 0 < 3 \Rightarrow 0 < 3 \text{ (True)}$$

So we shade the region **above** the line (the side containing $(0, 0)$).

Illustration:



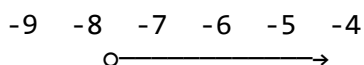
ROBUST WORKED EXAMPLES

Example 1: Solving Inequality

Solve and represent: ($3x - 5 < 4x + 2$)

$$[3x - 5 < 4x + 2] [-5 - 2 < 4x - 3x] [-7 < x] \text{ or } [x > -7]$$

Graph:

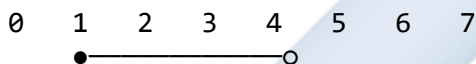


Example 2: Double Inequality

Solve and represent: ($-1 \leq 2x - 5 < 7$)

$$[-1 + 5 \leq 2x < 7 + 5] [4 \leq 2x < 12] [2 \leq x < 6]$$

Graph:



Example 3: Graphing Inequality

Graph the inequality ($y > -x + 1$).

Boundary line: ($y = -x + 1$)

- Slope = -1
- y-intercept = 1

Test (0,0): $0 > -0 + 1 \Rightarrow 0 > 1$ (False)

Shade the region **above** the line (not containing origin).

REVISION QUESTIONS

1. Solve for (x): ($3(x - 2) < 5x + 4$)
2. Represent ($-1 \leq 2x - 5 < 7$) on a number line.
3. State the “Golden Rule” in inequalities.
4. When do we use a dashed line vs. a solid line in graphs?
5. How do we determine which side of a line to shade?

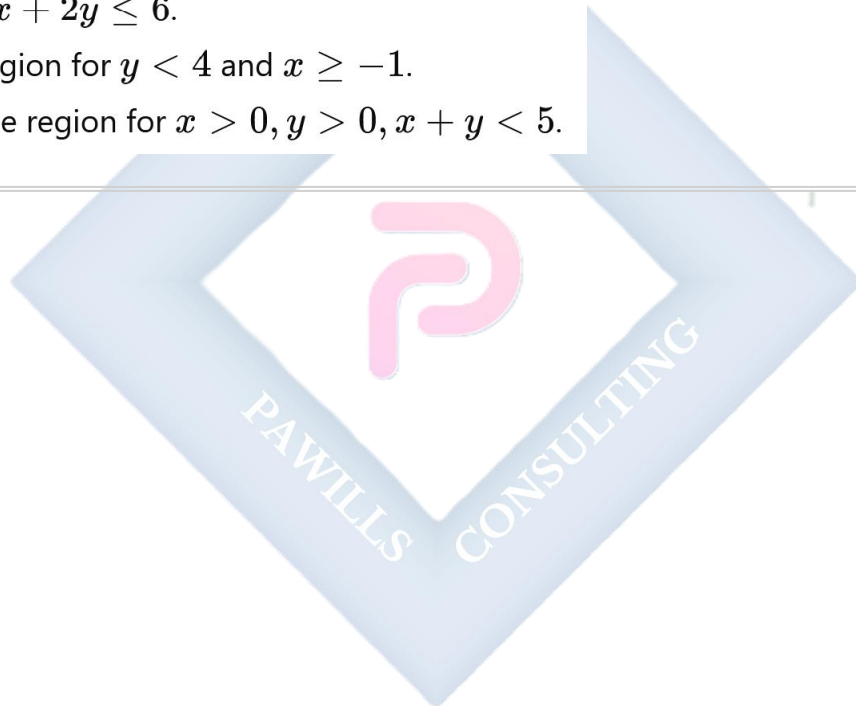
ASSESSMENT

EVALUATION (IN-CLASS)

1. Solve and represent $\frac{x}{2} - 5 > -8$
2. Solve $7 - 4x \leq 1$
3. Solve $-5 \leq 3x + 1 \leq 10$
4. Graph $y > -x + 1$
5. Graph $x \geq 3$ (vertical line through 3)

ASSIGNMENT (TAKE-HOME)

1. Solve $\frac{2x-1}{3} \geq \frac{x+5}{2}$
2. Find range of x for $3x - 2 > 7$ and $x - 1 < 5$.
3. Graph $3x + 2y \leq 6$.
4. Graph region for $y < 4$ and $x \geq -1$.
5. Shade the region for $x > 0, y > 0, x + y < 5$.



WEEK 8: BINARY OPERATIONS

TOPIC: BINARY OPERATIONS

INTRODUCTION

A **binary operation** is a rule that combines two elements to produce a single new element.

- "Binary" means *two inputs*.
- Common examples:
 - $(a + b)$
 - $(a - b)$
 - $(a \times b)$

We can invent our own binary operations using symbols like $*$, $\odot$, or Δ .

CONTENT BREAKDOWN

1. DEFINITION

If $(*)$ is a binary operation on a set (S) , then:

$[a * b \in S \text{ for all } a, b \in S]$

Example: If $(a * b = a + b + 2ab)$, then: $[3 * 5 = 3 + 5 + 2(3)(5) = 8 + 30 = 38]$

2. PROPERTIES (LAWS)

Property	Meaning	Example
Closure	$(a * b \in S \text{ for all } (a, b \in S))$	Integers under $+$
Commutative	$(a * b = b * a)$	$2 + 3 = 3 + 2$
Associative	$((a * b) * c = a * (b * c))$	$(2 + 3) + 4 = 2 + (3 + 4)$
Distributive	$(a * (b + c) = (a * b) + (a * c))$	For $\times$ over $+$

3. IDENTITY ELEMENT (e)

An element $(e \in S)$ is **identity** if: $[a * e = e * a = a]$

Examples:

- For $+$, ($e = 0$)
- For $\times$, ($e = 1$)

4. INVERSE ELEMENT (a^{-1})

The **inverse** of (a) is an element such that: $[a * a^{-1} = a^{-1} * a = e]$

Examples:

- For $+$, inverse of (a) is ($-a$).
- For $\times$, inverse of (a) is ($\frac{1}{a}$).

5. MULTIPLICATION (CAYLEY) TABLE

Used for **finite sets** to test closure, commutativity, identity, and inverses.

ILLUSTRATIVE TABLE

Operation (Δ) defined on ($S = \{1,2,3,4\}$): $[a \Delta b = (a \times b) \bmod 5]$

Δ	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

✓ **Closed** (all entries in S) ✓ **Commutative** (symmetric along diagonal) ✓ **Identity = 1** (first row/column matches header) ✓ **Inverses:** $2 \leftrightarrow 3$, $3 \leftrightarrow 2$, $4 \leftrightarrow 4$

ROBUST WORKED EXAMPLES

Example 1:

If ($a * b = a + b + ab$):

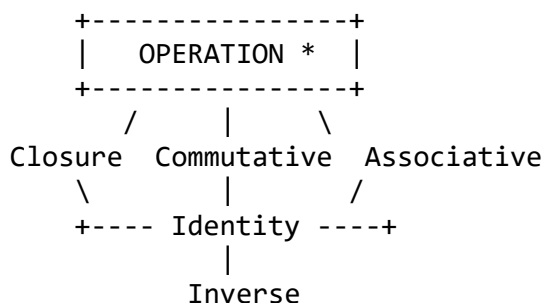
a) Is $*$ commutative? Yes, ($ab = ba$).

b) Identity? Solve ($a * e = a$): $[a + e + ae = a \Rightarrow e(1 + a) = 0 \Rightarrow e = 0]$

c) Inverse of 5? $5 * x = 0 \Rightarrow 5 + x + 5x = 0 \Rightarrow x(1 + 5) = -5 \Rightarrow x = -\frac{5}{6}$

GRAPHICAL SUMMARY

PROPERTIES OF BINARY OPERATIONS



REVISION QUESTIONS

1. If $(x*y = x^2 + y)$, is $*$ commutative? Associative?
2. Using Δ table above, find $((2\Delta 4)\Delta 3)$ and (4^{-1}) .
3. For $(a*b = a + b - 3)$, find identity (e) .
4. Using (e) from (3), find inverse of 10.

ASSESSMENT

EVALUATION (IN-CLASS)

1. $(ab = 2a - b)$. Find (35) and $(5*3)$. Is $*$ commutative?
2. From Δ table, solve $(3\Delta(n\Delta 2) = 4)$.
3. $(p\Delta q = p + q + 1)$. Find identity.
4. Find inverse of 7 using the operation in (3).
5. What does it mean for a set to be “closed” under $*$?

ASSIGNMENT (TAKE-HOME)

1. $(x\odot y = x + y - 2xy)$: check commutativity and associativity.
2. $(m*n = \frac{m+n}{2})$: find identity and inverse.
3. Using set T and operation table (given), find closure, commutativity, identity, and inverses.
4. Define $(a*b = a^b)$: check commutativity and associativity.
5. Let $(ab = a + 2b)$ and $(a\Delta b = 2a + b)$. Check distributivity: $(a(b\Delta c) = (ab)\Delta(ac))$?

WEEK 9: FUNCTIONS

TOPIC: FUNCTIONS

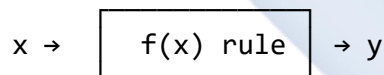
INTRODUCTION

In mathematics, a **function** is one of the most important foundational ideas. It defines a *relationship* between two quantities such that each input (x) corresponds to exactly one output (y).

You can imagine a **function** as a **machine**:

Input (x)	Function Machine	Output (y = f(x))
2	$\times 2 + 3$	7
5	$\times 2 + 3$	13
8	$\times 2 + 3$	19

Visually:



For every input value (x), the machine applies a *rule* (f) to produce an output (y).

CONTENT BREAKDOWN

1. DEFINITION OF A FUNCTION

A **function** (f) from a set **A** to a set **B** is a rule that assigns to each element (x) in **A** *exactly one* element (y) in **B**.

We write this as:

$$[f: A \rightarrow B]$$

and

$$[y = f(x)]$$

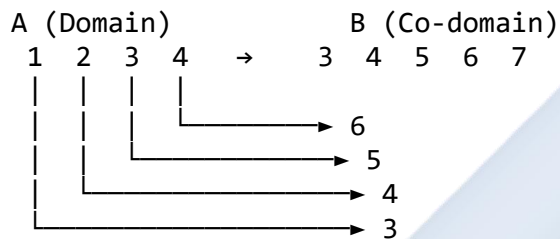
- **A** = *Domain* (the set of all possible input values)

- **B = Co-domain** (the set of all possible output values)
- **Range** = The actual outputs obtained after applying the function rule (subset of B)

Illustration:

Domain (A)	Rule	Co-domain (B)	Range
{1, 2, 3, 4}	$f(x) = x + 2$	{3, 4, 5, 6, 7}	{3, 4, 5, 6}

Mapping Diagram:



2. TYPES OF MAPPINGS (FUNCTIONS)

There are different ways to classify functions depending on how the domain and range are related.

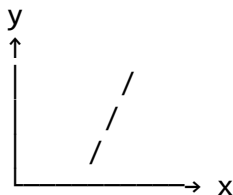
A. One-to-One (Injective) Function

Each element of the **domain** maps to a *unique* element of the **co-domain**. No two different x-values share the same y-value.

Example: ($f(x) = x + 1$)

x	f(x)
1	2
2	3
3	4

Graphically:



A straight line with a positive slope passes the **horizontal line test**, meaning it's one-to-one.

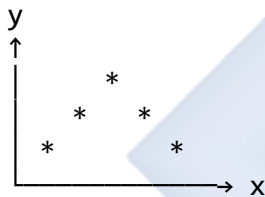
B. Many-to-One Function

Two or more x-values produce the same y-value.

Example: ($f(x) = x^2$)

x	f(x)
-2	4
-1	1
0	0
1	1
2	4

Graphically:



This fails the **horizontal line test**, so it's not one-to-one.

C. Onto (Surjective) Function

Every element in the **co-domain** has at least one pre-image in the **domain**. That is, the range = co-domain.

Example: ($f(x) = x - 1$), where domain = {2, 3, 4}, co-domain = {1, 2, 3}.

Range = {1, 2, 3} = co-domain $\Rightarrow$ **Onto function**

D. Into Function

The range is a **proper subset** of the co-domain — some elements in the co-domain have *no pre-image*.

Example: ($f(x) = 2x$), domain = {1, 2, 3}, co-domain = {1, 2, 3, 4, 5, 6}. Range = {2, 4, 6} $\subset$ Co-domain $\Rightarrow$ **Into function**

3. CLASSES AND TYPES OF FUNCTIONS

Function Type	Rule	Description	Example / Graph
Constant Function	$f(x) = c$	Always gives the same output	$f(x)=5 \rightarrow$ horizontal line
Identity Function	$f(x) = x$	Output equals input	Straight line $y=x$
Exponential Function	$f(x) = a^x$	Grows or decays rapidly	$y=2^x$
Logarithmic Function	$f(x) = \log_a(x)$	Inverse of exponential	$y=\log_2 x$
Trigonometric (Circular)	$f(x)=\sin x, \cos x$	Periodic wave-like behavior	$y=\sin x$
Composite Function	$(f \circ g)(x)=f(g(x))$	Function inside another	$f(x)=2x, g(x)=x+3 \rightarrow f(g(x))=2(x+3)=2x+6$
Inverse Function	$f^{-1}(x)$	Reverses $f(x)$	$f(x)=2x+3 \rightarrow f^{-1}(x)=(x-3)/2$

4. HOW TO FIND INVERSE AND COMPOSITE FUNCTIONS

A. Composite Function ($(f \circ g)(x)$)

Step 1: Start with $f(x)$.

Step 2: Replace every x in $f(x)$ with the expression for $g(x)$.

$$[(f \circ g)(x) = f(g(x))]$$

B. Inverse Function ($f^{-1}(x)$)

To find the inverse of ($f(x)$):

1. Write ($y = f(x)$)
2. Swap x and y
3. Solve for y
4. Replace y with ($f^{-1}(x)$)

WORKED EXAMPLES

Example 1: Composite Functions

Let ($f(x)=2x+1$) and ($g(x) = x^2 - 3$). Find: a) ($f(g(x))$) b) ($g(f(x))$) c) ($f(f(2))$)

Solution:

$$\text{a) } (f(g(x))) = 2(x^2-3) + 1 = 2x^2 - 6 + 1 = 2x^2 - 5$$

b) ($g(f(x)) = (2x+1)^2 - 3 = 4x^2 + 4x - 2$)

c) ($f(f(2)) = f(2(2)+1) = f(5) = 2(5) + 1 = 11$)

Example 2: Inverse Function

Find the inverse of $f(x) = \frac{3x-1}{x+5}$.

Solution:

Step 1: $y = \frac{3x-1}{x+5}$

Step 2: Swap x and y $\rightarrow x = \frac{3y-1}{y+5}$

Step 3: Multiply both sides by (y+5):

$$x(y+5) = 3y-1$$

$$xy + 5x = 3y - 1$$

$$xy - 3y = -5x - 1$$

$$y(x-3) = -5x-1$$

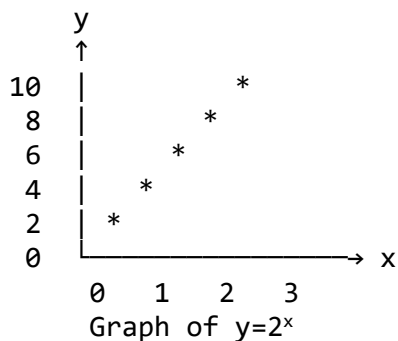
$$y = \frac{-5x-1}{x-3}$$

$$f^{-1}(x) = \frac{5x+1}{3-x}$$

Example 3: Graphical Representation

Function	Shape	Description
$f(x)=x$	Diagonal straight line	Passes through (0,0)
$f(x)=x^2$	Parabola	Opens upward
$f(x)=1/x$	Hyperbola	Approaches both axes asymptotically
$f(x)=2^x$	Exponential	Rises rapidly
$f(x)=\log x$	Logarithmic	Inverse of exponential

Illustration:



REVISION QUESTIONS

1. What is the difference between the **Co-domain** and **Range**?
2. Is $f(x) = |x|$ a one-to-one function? Explain.
3. If $f(x) = x + 2$ and $g(x) = 3x$, find $f(g(1))$ and $g(f(1))$.
4. Find the inverse of $f(x) = 2x - 7$.
5. What is the domain of $f(x) = \frac{1}{x-2}$?

ASSESSMENT

EVALUATION (IN-CLASS)

1. Define a **one-to-one** function.
2. What is a **constant** function?
3. Given $f(x) = 3x + 5$, find $f(2)$ and $f(-1)$.
4. If $h(x) = \frac{x+1}{2}$, find $h^{-1}(x)$.
5. If $f(x) = x^2$ and $g(x) = x + 1$, find $f(g(x))$.

ASSIGNMENT (TAKE-HOME)

1. Let $f(x) = 2x - 3$ and $g(x) = \frac{1}{x}$. Find:
a) $f(g(x))$ b) $g(f(x))$ c) $g(f(2))$.
2. Find the inverse of $g(x) = \frac{x+4}{2x-3}$.
3. A function $f(x) = x^2 + 1$:
a) Find its range.
b) Is it one-to-one? Explain.
c) Find $f(f(x))$.
4. If $f(x) = 3x - 1$, find $f^{-1}(x)$ and show that $(f \circ f^{-1})(5) = 5$.
5. Given $f(x) = 2x + 1$ and $g(x) = 3x - 1$, find x such that $f(g(x)) = g(f(x))$.

VISUAL SUMMARY

Concept	Symbol/Definition	Illustration
Function	$f: A \rightarrow B$	One input $\rightarrow$ one output
Inverse	$f^{-1}(x)$	Reverses $f(x)$
Composite	$(f \circ g)(x) = f(g(x))$	Combine functions
One-to-one	Each $x \rightarrow$ unique y	Passes horizontal line test
Onto	Range = Co-domain	All outputs used

WEEK 10: SEQUENCE AND SERIES

TOPIC: SEQUENCE AND SERIES

INTRODUCTION

In everyday life, we encounter **patterns of numbers** that follow a definite rule or order — such as house numbering, seat arrangements, or daily savings.

In mathematics, such arrangements are called **sequences**, and when the terms of a sequence are **added together**, the result is called a **series**.

CONTENT

1. DEFINITION OF TERMS

- **Sequence:** A sequence is an **ordered list of numbers** that follow a specific rule or pattern. Example: 2, 4, 6, 8, 10, ... (each term increases by 2).
 - **Term:** Each number in a sequence is called a **term**. The first term is denoted by **a**, and the n th term is denoted by T_n or U_n .
 - **Series:** When the terms of a sequence are **added**, we get a series. Example: $2 + 4 + 6 + 8 + \dots$
 - **Types of Sequences:**
 - a. Arithmetic Sequence (A.P.)
 - b. Geometric Sequence (G.P.)
 - c. Harmonic Sequence (H.P.)
-

2. ARITHMETIC PROGRESSION (A.P.)

An **Arithmetic Progression (A.P.)** is a sequence in which **each term differs from the previous one by a constant value**, called the **common difference (d)**.

Examples:

- 2, 4, 6, 8, 10, ... ($d = 2$)
- 100, 90, 80, 70, ... ($d = -10$)

General Form: [a, , $a + d$, , $a + 2d$, , $a + 3d$,]

FORMULAS

1. **nth Term (T_n)**

$$T_n = a + (n-1)d$$

2. **Sum of n Terms (S_n)**

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

or

$$S_n = \frac{n}{2}(a + l)$$

where l = last term.

ILLUSTRATION

Example Sequence: 3, 7, 11, 15, ...

Term	Value	Formula	Calculation
1st	3	a	3
2nd	$3 + d$	$3 + 4 = 7$	
3rd	$3 + 2d$	$3 + 8 = 11$	
4th	$3 + 3d$	$3 + 12 = 15$	

WORKED EXAMPLES

Example 1: Find the 15th term of the sequence 5, 8, 11, 14, ...

Solution: $a = 5$, $d = 3$

$$T_{15} = 5 + (15 - 1) \times 3 = 5 + 42 = 47$$

Answer: 47

Example 2: Find the sum of the first 10 terms of the sequence 2, 5, 8, 11, ...

Solution: $a = 2$, $d = 3$, $n = 10$

$$S_{10} = \frac{10}{2}[2(2) + (10-1) \times 3]$$

$$S_{10} = 5[4 + 27] = 5 \times 31 = 155$$

Answer: 155

3. GEOMETRIC PROGRESSION (G.P.)

A **Geometric Progression** is a sequence in which **each term after the first is obtained by multiplying the preceding term by a constant number**, called the **common ratio (r)**.

Examples:

- 2, 4, 8, 16, ... ($r = 2$)
- 81, 27, 9, 3, ... ($r = \frac{1}{3}$)

General Form: $a, ar, ar^2, ar^3, \dots$

FORMULAS

1. **nth Term (T_n)**

$$T_n = ar^{n-1}$$

2. **Sum of n Terms (S_n)**

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{for } r \neq 1$$

3. **Sum to Infinity (S_∞)**

When $|r| < 1$,

$$S_\infty = \frac{a}{1-r}$$

ILLUSTRATION

Example Sequence: 2, 4, 8, 16, ...

Term	Expression	Value
1st	a	2
2nd	ar	$2 \times 2 = 4$
3rd	ar^2	$2 \times 4 = 8$
4th	ar^3	$2 \times 8 = 16$

WORKED EXAMPLES

Example 1: Find the 5th term of the G.P. 3, 6, 12, 24, ...

Solution: $a = 3, r = 2$ [$T_5 = 3 \times 2^{5-1} = 3 \times 16 = 48$] **Answer: 48**

Example 2: Find the sum of the first 5 terms of the G.P. 2, 6, 18, 54, ...

Solution: $a = 2, r = 3, n = 5$ [$S_5 = \frac{2(3^5 - 1)}{3 - 1} = \frac{2(243 - 1)}{2} = 242$]
Answer: 242

Example 3: Find the sum to infinity of the G.P. 10, 5, 2.5, 1.25, ...

Solution: $a = 10, r = \frac{1}{2}$ (since $5/10 = \frac{1}{2}$) [$S_{\infty} = \frac{10}{1 - \frac{1}{2}} = \frac{10}{\frac{1}{2}} = 20$]
Answer: 20

4. HARMONIC PROGRESSION (H.P.)

A **Harmonic Progression** is a sequence whose **reciprocals form an arithmetic progression**.

If $[\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots]$ is an A.P., then $a, a+d, a+2d, \dots$ form a **H.P.**

Example: $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ is a harmonic progression.

5. RELATIONSHIP BETWEEN A.P., G.P., AND H.P.

Progression	Definition	General Term	Constant
Arithmetic	Each term differs by addition/subtraction	$T_n = a + (n-1)d$	Common Difference (d)
Geometric	Each term differs by multiplication/division	$T_n = ar^{n-1}$	Common Ratio (r)
Harmonic	Reciprocals form an A.P.	$T_n = 1 / [a + (n-1)d]$	Common Difference of reciprocals

GRAPHICAL ILLUSTRATIONS

A.P. (Linear growth)



G.P. (Exponential growth)



PRACTICAL APPLICATIONS

- A.P. → Used in salary increments, uniform savings, arithmetic coding.
- G.P. → Used in population growth, compound interest, depreciation.
- H.P. → Used in calculating average speeds and electrical resistances in parallel.

REVISION QUESTIONS

1. What is the difference between a sequence and a series?
2. Define the term “common ratio.”
3. Write down the first 5 terms of the sequence with $a = 4$, $d = 5$.
4. Find the 10th term of the G.P. whose $a = 2$ and $r = 3$.
5. State one real-life example of an arithmetic progression.

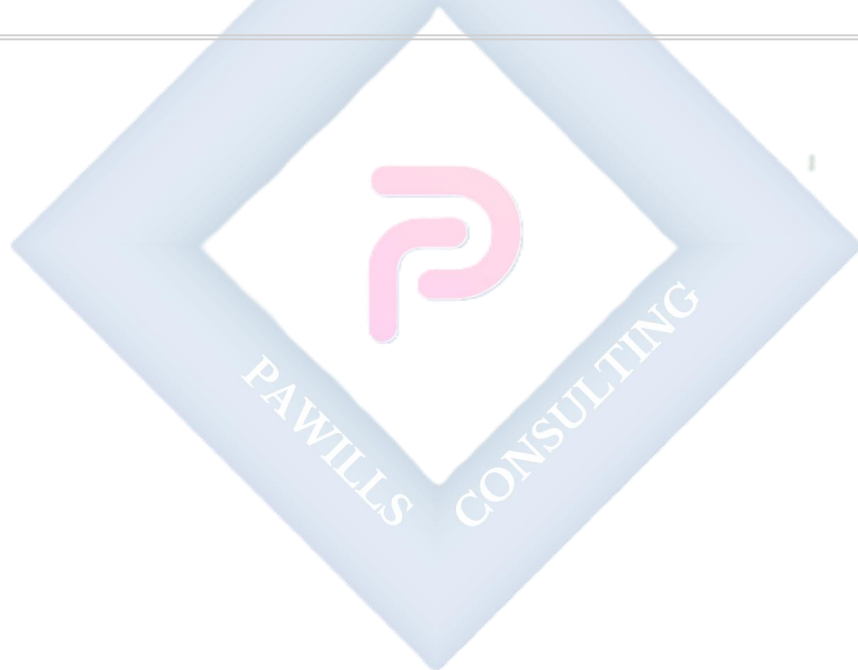
EVALUATION (IN-CLASS)

1. If the 1st term of an A.P. is 7 and the 5th term is 19, find the common difference.
2. Find the 20th term of the A.P. 4, 9, 14, ...
3. Find the sum of the first 6 terms of a G.P. with $a = 5$ and $r = 2$.
4. Find the sum to infinity of the G.P. 12, 8, $5\frac{1}{3}$, ...
5. If $1/a$, $1/b$, $1/c$ are in A.P., show that $b = 2ac / (a + c)$.

ASSIGNMENT (TAKE-HOME)

1. The 5th and 10th terms of an A.P. are 18 and 33 respectively.

- Find the first term and common difference.
- Find the sum of the first 15 terms.
- 2. The first term of a G.P. is 2, and its common ratio is $\frac{1}{2}$.
 - Find the sum to infinity.
 - Find the 8th term.
- 3. If the 3rd and 7th terms of a G.P. are 16 and 256, find:
 - (a) The common ratio
 - (b) The first term
 - (c) The 10th term
- 4. Write down the first 5 terms of the H.P. whose corresponding A.P. is 2, 4, 6, 8, 10.
- 5. Find the sum of all 3-digit numbers divisible by 9.



SS1 FURTHER MATHEMATICS LESSON NOTES

SECOND TERM

WEEK 1: TRIGONOMETRIC RATIOS OF SPECIAL ANGLES & LOGICAL REASONING (PART I)

TOPIC 1: TRIGONOMETRIC RATIOS OF 30° , 45° , AND 60°

INTRODUCTION

Trigonometry is a branch of mathematics that studies the relationships between the sides and angles of triangles. The three basic trigonometric ratios are:

- **Sine (sin) = Opposite / Hypotenuse**
- **Cosine (cos) = Adjacent / Hypotenuse**
- **Tangent (tan) = Opposite / Adjacent**

Among all possible angles, 30° , 45° , and 60° are called special angles because their trigonometric ratios have exact values that can be derived geometrically — without using a calculator.

These angles frequently appear in geometry, physics, navigation, and engineering, and understanding them gives a strong foundation for solving trigonometric problems.

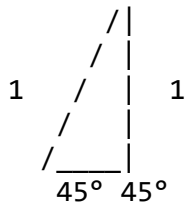
CONTENT

1. DERIVATION OF TRIGONOMETRIC RATIOS FOR 45°

We begin with an isosceles right-angled triangle, where the two non-right angles are equal — each 45° .

Let each of the equal sides be 1 unit.

Diagram:



Step 1: Use Pythagoras' Theorem to find the hypotenuse

$$h^2 = 1^2 + 1^2 = 2 \Rightarrow h = \sqrt{2}$$

Step 2: Apply SOH-CAH-TOA

- $\sin 45^\circ = \text{Opposite} / \text{Hypotenuse} = 1 / \sqrt{2} = \sqrt{2} / 2$
- $\cos 45^\circ = \text{Adjacent} / \text{Hypotenuse} = 1 / \sqrt{2} = \sqrt{2} / 2$
- $\tan 45^\circ = \text{Opposite} / \text{Adjacent} = 1 / 1 = 1$

✓ **Therefore:**

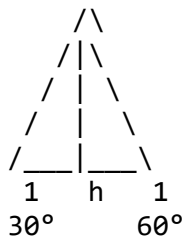
Ratio	Exact Value
$\sin 45^\circ$	$\sqrt{2}/2$
$\cos 45^\circ$	$\sqrt{2}/2$
$\tan 45^\circ$	1

2. DERIVATION OF TRIGONOMETRIC RATIOS FOR 30° AND 60°

Now, we derive from an equilateral triangle (all sides and all angles equal).

Let each side = 2 units, and each angle = 60° .

Step 1: Draw and bisect the triangle vertically



The bisector divides the 60° vertex into two 30° angles and splits the base (2 units) into 1 unit each.

Step 2: Find the height (h) using Pythagoras' theorem

$$2^2 = 1^2 + h^2 \Rightarrow 4 = 1 + h^2 \Rightarrow h^2 = 3 \Rightarrow h = \sqrt{3}$$

Step 3: Apply SOH-CAH-TOA

For the 60° angle:

- $\sin 60^\circ = \text{Opposite} / \text{Hypotenuse} = \sqrt{3} / 2$
- $\cos 60^\circ = \text{Adjacent} / \text{Hypotenuse} = 1 / 2$
- $\tan 60^\circ = \text{Opposite} / \text{Adjacent} = \sqrt{3} / 1 = \sqrt{3}$

For the 30° angle:

- $\sin 30^\circ = \text{Opposite} / \text{Hypotenuse} = 1 / 2$
- $\cos 30^\circ = \text{Adjacent} / \text{Hypotenuse} = \sqrt{3} / 2$
- $\tan 30^\circ = \text{Opposite} / \text{Adjacent} = 1 / \sqrt{3} = \sqrt{3} / 3$

3. SUMMARY TABLE OF SPECIAL ANGLES

Angle (θ)	$\sin \theta$	$\cos \theta$	$\tan \theta$
30°	1/2	$\sqrt{3}/2$	$\sqrt{3}/3$ (or $1/\sqrt{3}$)
45°	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60°	$\sqrt{3}/2$	1/2	$\sqrt{3}$

4. RECIPROCAL TRIGONOMETRIC RATIOS

Every trigonometric ratio has a reciprocal (or inverse) ratio:

Function	Reciprocal Function	Definition	Example
$\sin \theta$	$\csc \theta$	$\csc \theta = 1/\sin \theta$	$\csc 30^\circ = 1/(1/2) = 2$
$\cos \theta$	$\sec \theta$	$\sec \theta = 1/\cos \theta$	$\sec 60^\circ = 1/(1/2) = 2$
$\tan \theta$	$\cot \theta$	$\cot \theta = 1/\tan \theta$	$\cot 45^\circ = 1/1 = 1$

ROBUST WORKED EXAMPLES

Example 1: Evaluating Expressions

Find the exact value of $(\sin 60^\circ + \cos 60^\circ) / \tan 45^\circ$

Solution:

Substitute the values:

$$(\sqrt{3}/2 + 1/2) / 1$$

Simplify numerator:

$$(\sqrt{3} + 1)/2$$

✓ **Answer:** $(\sqrt{3} + 1) / 2$

Example 2: Solving for an Unknown

Find x if $x \sin 30^\circ \cos 45^\circ = \tan 60^\circ$.

Solution:

Substitute known values:

$$x \left(\frac{1}{2}\right) \left(\frac{\sqrt{2}}{2}\right) = \sqrt{3}$$

Simplify:

$$x \left(\frac{\sqrt{2}}{4}\right) = \sqrt{3}$$

Multiply both sides by $(4/\sqrt{2})$:

$$x = (4\sqrt{3})/\sqrt{2}$$

Rationalize denominator:

$$x = (4\sqrt{3} \times \sqrt{2})/2 = 2\sqrt{6}$$

✓ **Answer:** $x = 2\sqrt{6}$

Example 3: Application in Geometry

In $\triangle ABC$, $\angle B = 90^\circ$, $\angle A = 30^\circ$, and $AC = 10$ cm. Find the exact lengths of AB and BC .

Solution:

We know:

- $\sin 30^\circ = \text{Opp/Hyp} = BC / 10$
- $\cos 30^\circ = \text{Adj/Hyp} = AB / 10$
- $\sin 30^\circ = \frac{1}{2} \Rightarrow \frac{1}{2} = BC / 10 \Rightarrow BC = 5$ cm
- $\cos 30^\circ = \frac{\sqrt{3}}{2} \Rightarrow \frac{\sqrt{3}}{2} = AB / 10 \Rightarrow AB = 5\sqrt{3}$ cm

✓ **Answer:** $BC = 5$ cm, $AB = 5\sqrt{3}$ cm

Example 4: Using Reciprocal Ratios

Find $\sec 30^\circ \times \csc 60^\circ$.

Solution:

- $\sec 30^\circ = 1 / \cos 30^\circ = 1 / (\sqrt{3}/2) = 2 / \sqrt{3}$
- $\csc 60^\circ = 1 / \sin 60^\circ = 1 / (\sqrt{3}/2) = 2 / \sqrt{3}$

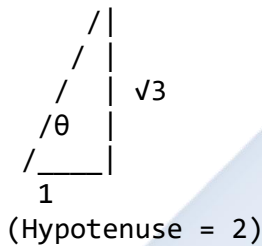
Multiply:

$$(2 / \sqrt{3}) \times (2 / \sqrt{3}) = 4 / 3$$

✓ **Answer:** 4/3

VISUAL RELATIONSHIPS OF SPECIAL ANGLES

$$\begin{aligned}\sin 30^\circ &= 1/2 \\ \cos 30^\circ &= \sqrt{3}/2\end{aligned}$$



KEY CONCEPTS TO REMEMBER

- The sum of squares identity always holds: $\sin^2 \theta + \cos^2 \theta = 1$
- $\tan \theta = \sin \theta / \cos \theta$
- The reciprocal identities: $\csc \theta = 1/\sin \theta$, $\sec \theta = 1/\cos \theta$, $\cot \theta = 1/\tan \theta$

EVALUATION (IN-CLASS)

1. Draw a 30° – 60° – 90° triangle and derive $\sin 60^\circ$ and $\tan 30^\circ$.
2. Find the exact value of $\sin 45^\circ + \cos 45^\circ$.
3. Compute $(\tan 60^\circ)^2 + (\sin 30^\circ)^2$.
4. Find the value of $\sec 30^\circ$.
5. If $2y \cos 30^\circ = \tan 60^\circ$, find y exactly.

ASSIGNMENT (TAKE-HOME)

1. Evaluate: $(\sin 60^\circ \cos 30^\circ) / (\tan 45^\circ - \sin 30^\circ)$
2. A ladder leans against a wall at 60° . If the base is 4 m from the wall, find the exact length of the ladder. (Hint: Use $\cos 60^\circ = \text{Adj}/\text{Hyp}$)
3. Solve for θ (acute): $2 \cos \theta = 1$

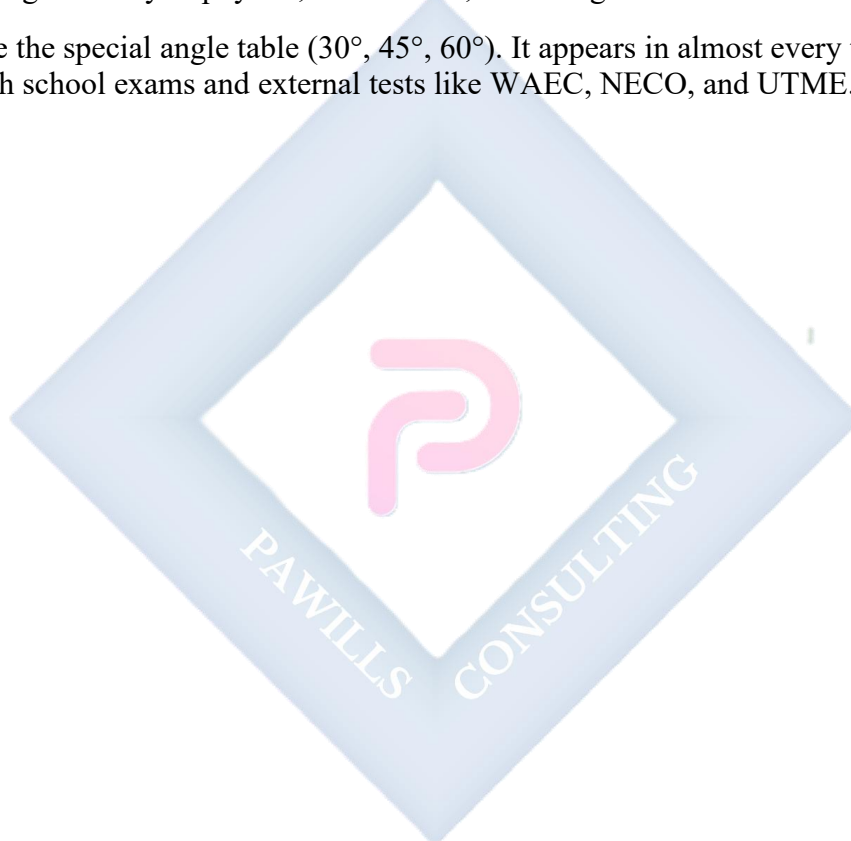
4. Find the exact value of $1 + \tan^2 60^\circ$ (show that it equals $\sec^2 60^\circ$).
5. Without using a calculator, verify: $\sin^2 30^\circ + \cos^2 30^\circ = 1$

CONCLUSION

Understanding these special angles forms the foundation of trigonometry. By mastering their geometric derivations and exact values, you can:

- Solve triangles without calculators
- Simplify trigonometric equations
- Apply trigonometry in physics, architecture, and navigation

Tip: Memorize the special angle table (30° , 45° , 60°). It appears in almost every trigonometric problem in both school exams and external tests like WAEC, NECO, and UTME.



WEEK 2: TRIGONOMETRIC RATIOS & LOGICAL REASONING (PART II)

TOPIC: INTRODUCTION TO LOGICAL REASONING

INTRODUCTION

Logical reasoning is the backbone of mathematical thinking. It allows us to analyze arguments, evaluate truth, and form valid conclusions. In mathematics, we often deal with statements (called propositions) that can be either true (T) or false (F).

Logical reasoning helps us understand how the truth of one statement affects another, and how we can build compound statements using logical connectives like and ($\wedge$), or ($\vee$), not ($\sim$), if...then ($\rightarrow$), and if and only if ($\leftrightarrow$).

This concept is not only useful in mathematics but also in computer programming, engineering, and critical thinking in everyday life.

CONTENT

1. PROPOSITIONS (STATEMENTS)

A proposition (or statement) is a declarative sentence that can be clearly classified as either True (T) or False (F), but not both.

Example	Statement	Type	Truth Value
p	Lagos is the capital of Nigeria.	Proposition	False
q	$2 + 2 = 4$	Proposition	True
r	What is your name?	Not a proposition	—
s	Go and shut the door.	Not a proposition	—
t	$x + 1 = 5$	Open sentence (depends on x)	—

A proposition must be definite — its truth value should not depend on unknown variables.

Visual Representation

SENTENCES IN LANGUAGE

- | |
|--|
| 1. Questions → Not Propositions
2. Commands → Not Propositions
3. Declarative → Propositions |
|--|

2. LOGICAL CONNECTIVES AND THEIR TRUTH TABLES

Logical connectives are symbols that join simple propositions to form compound propositions. We'll explore five main logical connectives and their corresponding truth tables.

A. NEGATION ($\sim p$ or $\neg p$)

This simply reverses the truth value of a proposition. If p is True, then $\sim p$ is False, and vice versa.

p	$\sim p$
T	F
F	T

Example: p : "It is sunny." → $\sim p$: "It is not sunny."

B. CONJUNCTION ($p \wedge q$)

The conjunction " p and q " is True only when both p and q are True.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example: p : "I passed the test." q : "I studied hard." $p \wedge q$ means "I passed the test and I studied hard."

C. DISJUNCTION ($p \vee q$)

This is the inclusive "or", meaning $p \vee q$ is True if either p or q is True, or both.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example: p: "I will travel to Lagos." q: "I will travel to Abuja." $p \vee q$ means "I will travel to Lagos or Abuja (or both)."

D. CONDITIONAL ($p \rightarrow q$)

This represents "if p, then q." It is False only when p is True and q is False.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Explanation (with analogy): "If I get an A (p), I will buy you ice cream (q)." I only lie (the statement is False) if I get an A but do not buy you ice cream.

E. BICONDITIONAL ($p \leftrightarrow q$)

This represents "p if and only if q." It is True only when p and q have the same truth value.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example: p: "Two angles are equal." q: "The triangle is isosceles." $p \leftrightarrow q$ means the triangle is isosceles if and only if two angles are equal.

3. TAUTOLOGY, CONTRADICTION, AND CONTINGENCY

Type	Definition	Example	Truth Value
Tautology	Always True, no matter what p or q are.	$p \vee \sim p$	Always T

Type	Definition	Example	Truth Value
Contradiction	Always False, no matter what p or q are.	$p \wedge \sim p$	Always F
Contingency	Sometimes True, sometimes False.	$p \rightarrow q$	Depends on p and q

Diagrammatic View

LOGICAL STATEMENTS
• Tautology $\rightarrow$ Always True • Contradiction $\rightarrow$ Always False • Contingency $\rightarrow$ Sometimes True

ROBUST WORKED EXAMPLES

Example 1: Translating to Symbols

Let p = "It is cold" q = "It is raining"

Translate:

a) It is cold and it is raining $\rightarrow p \wedge q$ b) It is not cold or it is raining $\rightarrow \sim p \vee q$ c) If it is raining, then it is not cold $\rightarrow q \rightarrow \sim p$ d) It is neither cold nor raining $\rightarrow \sim p \wedge \sim q$

Example 2: Constructing a Truth Table

Construct a truth table for: $(p \rightarrow q) \wedge \sim q$

p	q	$\sim q$	$p \rightarrow q$	$(p \rightarrow q) \wedge \sim q$
T	T	F	T	F
T	F	T	F	F
F	T	F	T	F
F	F	T	T	T

✓ Final column is not constant, so it's a contingency.

Example 3: Checking for Tautology

Is $(p \wedge q) \rightarrow p$ a tautology?

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
---	---	--------------	------------------------------

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

✓ The last column is always True $\rightarrow$ Tautology

Example 4: Identifying a Contradiction

Check if $(p \wedge \sim p)$ is a contradiction.

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

✓ Always False $\rightarrow$ Contradiction

Example 5: Logical Reasoning in Real Life

Statement: "If a number is even, then it is divisible by 2."

Let p: "A number is even." Let q: "It is divisible by 2." $\rightarrow$ This can be written as $p \rightarrow q$

Truth Table confirms it's a tautology, because it's always true by definition.

VISUAL SUMMARY

LOGICAL CONNECTIVES MAP

Simple Statements (p, q)

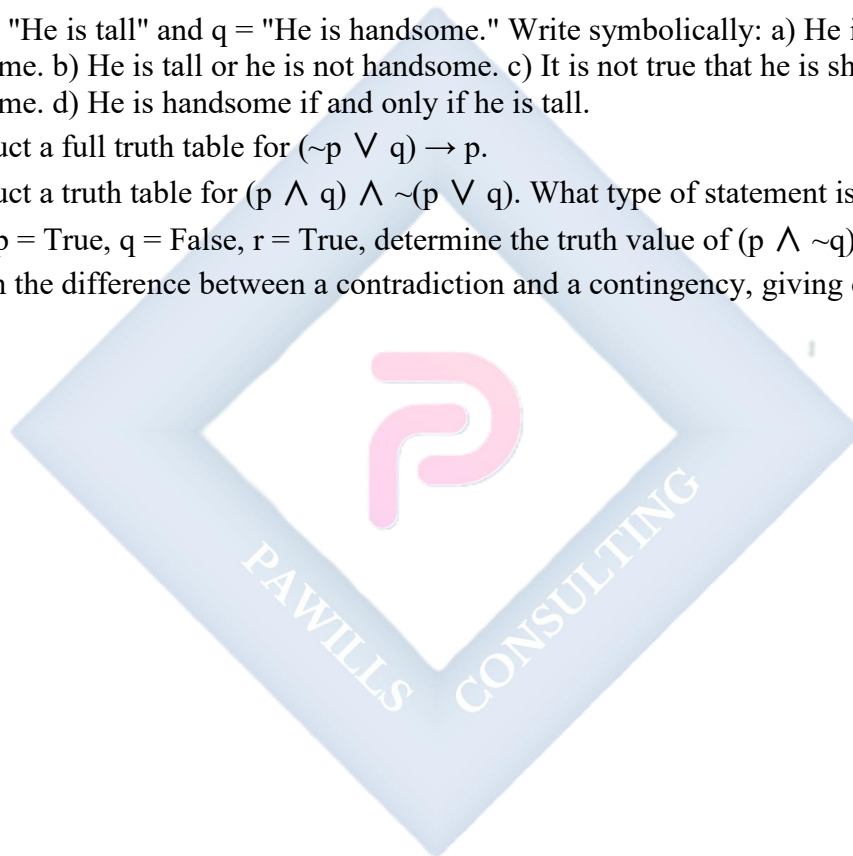
- Negation ($\sim p$) $\rightarrow$ reverses truth
- Conjunction ($p \wedge q$) $\rightarrow$ both true
- Disjunction ($p \vee q$) $\rightarrow$ either true
- Conditional ($p \rightarrow q$) $\rightarrow$ if...then
- Biconditional ($p \leftrightarrow q$) $\rightarrow$ both same

EVALUATION (IN-CLASS)

1. Which of the following are propositions? a) $x > 5$ b) $10 - 4 = 6$ c) Where are you going?
 2. Let p be "The sun is hot" and q be "The moon is made of cheese." Find the truth value of $p \wedge q$.
 3. What is the only condition under which $p \rightarrow q$ is False?
 4. What is the negation of: r : "I am 15 years old"?
 5. Construct a truth table for $p \vee \sim p$. What kind of statement is this?
-

ASSIGNMENT (TAKE-HOME)

1. Let p = "He is tall" and q = "He is handsome." Write symbolically: a) He is tall and handsome. b) He is tall or he is not handsome. c) It is not true that he is short and handsome. d) He is handsome if and only if he is tall.
2. Construct a full truth table for $(\sim p \vee q) \rightarrow p$.
3. Construct a truth table for $(p \wedge q) \wedge \sim(p \vee q)$. What type of statement is this?
4. Given p = True, q = False, r = True, determine the truth value of $(p \wedge \sim q) \rightarrow (r \vee q)$.
5. Explain the difference between a contradiction and a contingency, giving one example each.



WEEK 3: CALCULATING AND PROCESSING DEVICES

INTRODUCTION

From the dawn of civilization, humans have sought ways to simplify calculation and data processing. Before the invention of modern computers, people created devices such as the Abacus and later, mechanical calculators, to handle complex arithmetic operations efficiently. In this lesson, we explore:

- The Abacus, one of the earliest calculating devices.
 - The Decimal (Base 10) and Binary (Base 2) number systems.
 - Conversions between number systems.
 - The use of Flow Charts for planning and problem-solving.
-

CONTENT BREAKDOWN

1. ABACUS (A CALCULATING DEVICE)

Definition:

An Abacus is a simple, manual calculating tool consisting of a frame with rods or wires, on which beads slide back and forth. It is used for performing arithmetic operations like addition, subtraction, multiplication, and division.

Structure:

An Abacus typically consists of:

- A Rectangular Frame
- Several Vertical Rods or Wires
- Beads threaded on the rods
- A Divider Bar separating place values (e.g., units, tens, hundreds)

Illustration:

o	o	o	o	o	o	← Upper deck (represents 5s)
o	o	o	o	o	o	← Lower deck (represents 1s)
o	o	o	o	o	o	
o	o	o	o	o	o	

(Each column represents a place value: ones, tens, hundreds, etc.)

Functions:

- It stores data temporarily by representing numbers as bead positions.
- It allows fast arithmetic through manual manipulation.
- It is used to teach counting and number systems.

Example:

If one bead in the lower deck equals 1, and one bead in the upper deck equals 5, then:

- Lower deck beads = 3
- Upper deck bead = 1 Total = 8 (5 + 3)

Modern Relevance:

Though primitive, the Abacus inspired the design of early mechanical calculators and shaped the evolution of computer memory concepts, where beads act like bits—each representing a unit of data.

2. DECIMAL (BASE 10) AND BINARY (BASE 2) SYSTEMS

A. Decimal System (Base 10):

- The Decimal system is the most familiar and widely used number system.
- It uses ten digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.
- Each digit's position has a place value that is a power of 10.

Example:

$$345_{10} = (3 \times 10^2) + (4 \times 10^1) + (5 \times 10^0) = 300 + 40 + 5 = 345$$

B. Binary System (Base 2):

- The Binary system uses only two digits: 0 and 1.
- It is the language of computers, where 0 represents "OFF" and 1 represents "ON".
- Each position is a power of 2.

Example:

$$1101_2 = (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) = 8 + 4 + 0 + 1 = 13_{10}$$

Illustration (Place Value Table):

Binary Digit	1	1	0	1
Power of 2	2^3	2^2	2^1	2^0
Value	8	4	0	1

Binary Digit	1	1	0	1
Sum				13

3. CONVERSIONS BETWEEN NUMBER SYSTEMS

A. Binary $\rightarrow$ Decimal

Steps:

1. Write the binary number.
2. Multiply each digit by its corresponding power of 2.
3. Add up all the results.

Example: Convert 10110_2 to decimal.

$$= (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) = 16 + 0 + 4 + 2 + 0 = 22_{10}$$

B. Decimal $\rightarrow$ Binary

Steps:

1. Divide the decimal number by 2.
2. Record the remainder.
3. Continue dividing the quotient by 2 until it equals 0.
4. Read the remainders from bottom to top.

Example: Convert 25_{10} to binary.

Division	Quotient	Remainder
$25 \div 2$	12	1
$12 \div 2$	6	0
$6 \div 2$	3	0
$3 \div 2$	1	1
$1 \div 2$	0	1

Binary Result: $25_{10} = 11001_2$

C. Binary Addition

Binary addition follows these rules:

1st Bit	2nd Bit	Sum	Carry
0	0	0	0

1st Bit	2nd Bit	Sum	Carry
0	1	1	0
1	0	1	0
1	1	0	1

Example:

$$1011_2 + 110_2$$

Align digits:

$$\begin{array}{r}
 1011 \\
 + 0110 \\
 \hline
 10001
 \end{array}$$

Therefore, $1011_2 + 110_2 = 10001_2 = 17_{10}$.

4. FLOW CHARTS

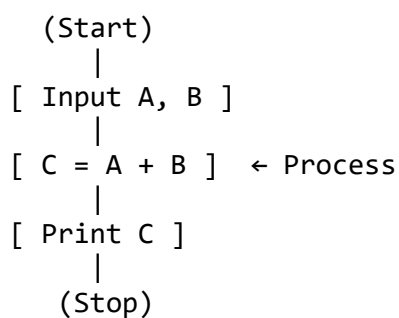
Definition:

A flow chart is a diagrammatic representation of the logical steps needed to solve a problem. It shows the sequence of operations using symbols connected by arrows.

Common Flow Chart Symbols:

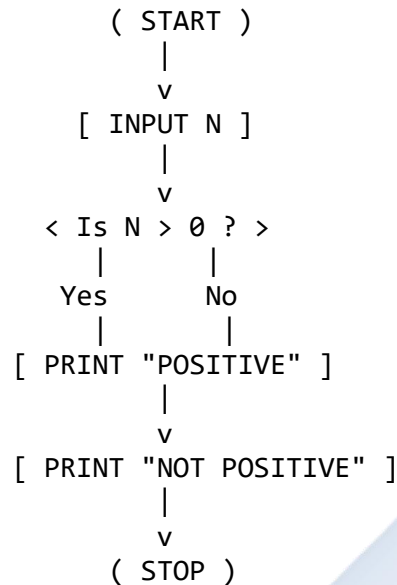
Symbol	Shape	Function
Oval	Terminal	Start or End of the process
■ Rectangle	Process	Indicates calculation or operation
▭ Parallelogram	Input/Output	Shows data entry or display
Diamond	Decision	Represents a Yes/No question
Arrow	Flow Line	Indicates direction of process flow

Illustration of Symbols:



Flow Chart Example:

Problem: Draw a flow chart to read a number (N) and print "POSITIVE" if $N > 0$, otherwise print "NOT POSITIVE".



ROBUST WORKED EXAMPLES

1. Convert 10110_2 to Decimal $\rightarrow 22_{10}$
 2. Convert 25_{10} to Binary $\rightarrow 11001_2$
 3. Add Binary Numbers: $1011_2 + 110_2 = 10001_2$
 4. Flow Chart Problem: Print whether a number is positive or not (shown above).
-

EVALUATION (IN-CLASS)

1. What is an Abacus, and how does it function?
 2. Differentiate between the Decimal and Binary systems.
 3. Convert 19_{10} to Binary.
 4. Convert 1001_2 to Decimal.
 5. List and explain three flow chart symbols and their functions.
-

ASSIGNMENT (TAKE-HOME)

1. Convert 11101_2 to Decimal.
2. Convert 45_{10} to Binary.
3. Add the binary numbers $1011_2 + 110_2$ and verify your answer by converting all to Decimal.
4. Draw a flow chart that takes two numbers (A) and (B) and prints the larger one.

5. Draw a flow chart to calculate and print the Area (A) of a rectangle given its Length (L) and Width (W).



WEEK 4: THE STRAIGHT LINE IN COORDINATE GEOMETRY

INTRODUCTION

Coordinate Geometry — also called Analytical Geometry — forms the powerful link between Algebra and Geometry. In this branch of mathematics, points, lines, and shapes are described using algebraic equations on a Cartesian plane, allowing us to study geometric figures numerically.

The straight line is the simplest and most fundamental of all geometric forms. Understanding its properties — midpoint, distance, gradient, and equation — is essential for solving higher problems in geometry, trigonometry, and calculus.

CONTENT

1. MIDPOINT OF A LINE SEGMENT

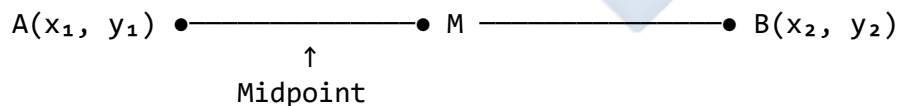
Concept:

The midpoint of a line segment joining two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is the point exactly halfway between them.

Formula:

$$M((x_1 + x_2)/2, (y_1 + y_2)/2)$$

Illustration:



The midpoint simply represents the average of the x-coordinates and the average of the y-coordinates.

2. GRADIENT (SLOPE) OF A STRAIGHT LINE

Concept:

The gradient (m) of a line measures its steepness and direction. It represents the ratio of the vertical change (Rise) to the horizontal change (Run) between any two points on the line.

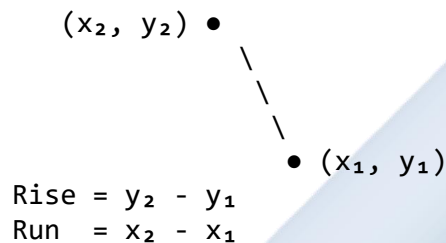
Formula:

$$m = \Delta y / \Delta x = (y_2 - y_1) / (x_2 - x_1)$$

Interpretation:

- A positive gradient ($m > 0$) → line rises from left to right.
- A negative gradient ($m < 0$) → line falls from left to right.
- $m = 0$ → horizontal line.
- Undefined m → vertical line.

Diagram:



3. DISTANCE BETWEEN TWO POINTS

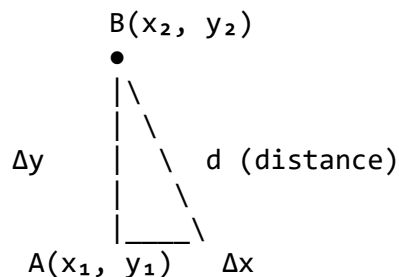
Concept:

The distance between two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is derived from Pythagoras' Theorem.

Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Illustration:



Here, $\Delta x = x_2 - x_1$ and $\Delta y = y_2 - y_1$.

4. CONDITIONS FOR PARALLELISM AND PERPENDICULARITY

A. Parallel Lines

Two lines are parallel if and only if their gradients are equal.

$$L_1 \parallel L_2 \Rightarrow m_1 = m_2$$

Parallel lines never meet.

B. Perpendicular Lines

Two lines are perpendicular if and only if the product of their gradients is -1.

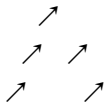
$$L_1 \perp L_2 \Rightarrow m_1 \times m_2 = -1$$

or equivalently,

$$m_2 = -1/m_1$$

Illustration:

Parallel Lines:



Perpendicular Lines:



5. EQUATION OF A STRAIGHT LINE

A. Gradient–Intercept Form

$$y = mx + c$$

where m = gradient, c = y-intercept (the point where the line cuts the y-axis).

Example: $y = 2x + 3$ means the line crosses the y-axis at $(0, 3)$ and has gradient 2.

B. Point–Gradient Form

$$y - y_1 = m(x - x_1)$$

Used when the gradient and one point (x_1, y_1) are known.

C. General Form

$$Ax + By + C = 0$$

All terms are on one side of the equation.

Example: $2x - y - 7 = 0$ can be rewritten as $y = 2x - 7$.

6. TRANSFORMING RELATIONS INTO LINEAR FORM

Purpose:

In experiments, some non-linear relationships (e.g., $y = ax^n$ or $y = ab^x$) can be linearized using logarithms. When converted to straight-line form, constants such as a , b , and n can be found from graphs.

Case 1: $y = ax^n$

Take logs of both sides:

$$\log y = \log a + n \log x$$

Comparing with $Y = mX + C$:

$$Y = \log y, X = \log x, m = n, C = \log a$$

Thus, a plot of $\log y$ (vertical axis) against $\log x$ (horizontal axis) gives a straight line with:

- Slope = n
 - Intercept = $\log a$
-

Case 2: $y = ab^x$

Take logs of both sides:

$$\log y = \log a + x \log b$$

Comparing with $Y = mX + C$:

$$Y = \log y, X = x, m = \log b, C = \log a$$

Thus, a plot of $\log y$ (Y-axis) against x (X-axis) gives a straight line.

7. AREA OF TRIANGLES AND QUADRILATERALS (SHOELACE METHOD)

Concept:

The area of a polygon whose coordinates are known can be found using the Shoelace Formula (also called Determinant Formula).

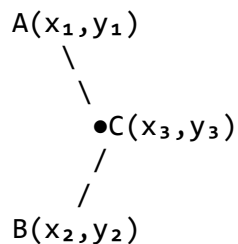
For a Triangle:

Given points $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$,

$$\text{Area} = \frac{1}{2} |(x_1y_2 + x_2y_3 + x_3y_1) - (y_1x_2 + y_2x_3 + y_3x_1)|$$

Diagrammatic Representation:

Vertices connected in order:



For a Quadrilateral:

Divide the shape into two triangles, calculate their individual areas, and add them together.

ROBUST WORKED EXAMPLES

Example 1: Midpoint, Distance, and Gradient

Given: P(-2,5) and Q(4,1)

Find: (a) Midpoint (M), (b) Distance (d), (c) Gradient (m)

Solution:

Let $(x_1, y_1) = (-2, 5)$, $(x_2, y_2) = (4, 1)$

(a) $M = ((-2 + 4)/2, (5 + 1)/2) = (1, 3)$

(b) $d = \sqrt{[(4 - (-2))]^2 + (1 - 5)^2} = \sqrt{6^2 + (-4)^2} = \sqrt{52} = 2\sqrt{13}$

(c) $m = (1 - 5)/(4 - (-2)) = -4/6 = -2/3$

Example 2: Parallel and Perpendicular Lines

Find the equation of a line passing through (3, -1) and perpendicular to $2y + x = 5$.

Solution:

$$2y + x = 5 \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$$

Gradient of given line $m_1 = -\frac{1}{2}$

Perpendicular line $\rightarrow m_2 = -1/m_1 = 2$

Use point-gradient form:

$$y - (-1) = 2(x - 3)$$

$$y + 1 = 2x - 6$$

$$y = 2x - 7 \text{ or } 2x - y - 7 = 0$$

Example 3: Area of a Triangle (Shoelace Formula)

Given: A(1,1), B(4,5), C(6,2)

$$\text{Area} = \frac{1}{2} |(1 \times 5 + 4 \times 2 + 6 \times 1) - (1 \times 4 + 5 \times 6 + 2 \times 1)|$$

$$= \frac{1}{2} |(5 + 8 + 6) - (4 + 30 + 2)|$$

$$= \frac{1}{2} |19 - 36| = \frac{1}{2} \times 17 = 8.5$$

Answer: Area = 8.5 square units

EVALUATION (IN-CLASS)

1. Find the distance between A(1, -2) and B(4, 2).
2. Find the midpoint of the line segment joining P(-3, 0) and Q(5, -6).
3. Find the gradient of the line passing through (1, 7) and (3, 1).
4. A line has the equation $3x - y = 6$. Find its gradient and y-intercept.
5. Are the lines $y = 2x + 1$ and $4x - 2y + 5 = 0$ parallel? Justify your answer.

ASSIGNMENT (TAKE-HOME)

1. A line L_1 passes through (2,5) and (4,9). A line L_2 passes through (0,1) and (-4,3). Determine if L_1 and L_2 are parallel, perpendicular, or neither.
2. Find the equation of the line parallel to $y = 3x - 1$ that passes through (2, 0).
3. Find the equation of the perpendicular bisector of the line joining A(-1,4) and B(3,2).
4. The variables x and y are related by $y = kx^n$. The table shows:

x	1	4	9
y	5	10	15

Explain how to plot a straight-line graph to find k and n . (What would you plot on the X and Y axes?)

5. Find the area of the quadrilateral with vertices P(1,2), Q(3,5), R(6,6), S(4,1).

WEEK 6: VECTORS IN TWO DIMENSIONS I

INTRODUCTION

In mathematics and science, quantities are described either by their size (magnitude) alone or by both magnitude and direction.

- Quantities with magnitude only are called **Scalars**.
- Quantities with magnitude and direction are called **Vectors**.

Vectors are used to represent forces, velocities, displacements, and many other physical quantities that involve direction. In two dimensions (2D), a vector lies on a flat plane, described by an x-axis (horizontal) and a y-axis (vertical).

CONTENT BREAKDOWN

1. SCALARS AND VECTORS

Quantity Type	Description	Examples
Scalar	Has magnitude only (no direction).	Distance (5 km), Time (30 s), Mass (10 kg), Speed (60 km/h)
Vector	Has magnitude and direction.	Displacement (5 km North), Force (10 N East), Velocity (80 km/h West)

Visual Illustration:

SCALAR (magnitude only) $\rightarrow$ 5 km

VECTOR (magnitude + direction) $\rightarrow$ 5 km North $\rightarrow$

In diagrams, a vector is represented by a directed line segment with:

- Length proportional to its magnitude.
- Arrowhead showing its direction.

2. VECTOR NOTATION

Vectors can be written in different but equivalent forms:

- **Geometric Form:** $\overrightarrow{AB}$ represents a vector from point A to point B.

- **Algebraic (Component) Form:** A vector from the origin to point (x, y) is written as

$$\vec{a} = (x) (y)$$

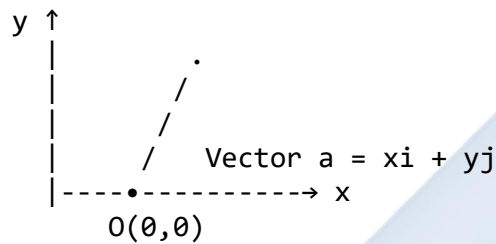
- **Unit Vector Form:**

$$\vec{a} = xi + yj$$

where $i = (1) \rightarrow$ unit vector along x-axis (0)

$j = (0) \rightarrow$ unit vector along y-axis
(1)

Illustration:



3. ZERO AND NEGATIVE VECTORS

- **Zero Vector ($\vec{0}$):**

$$\vec{0} = (0) (0)$$

It has zero magnitude and no direction.

- **Negative Vector ($-\vec{a}$):** Has the same magnitude as $\vec{a}$ but opposite direction.

If $\vec{a} = (x) (y)$, then $-\vec{a} = (-x) (-y)$

Visual Concept:

$$\begin{aligned} &\rightarrow \vec{a} \\ &\leftarrow -\vec{a} \end{aligned}$$

4. VECTOR ADDITION AND SUBTRACTION

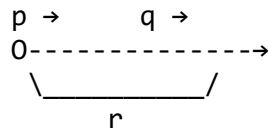
Vectors are added or subtracted component-wise.

$$(a) + (c) = (a + c) (b) (d) (b + d)$$

$$(a) - (c) = (a - c) (b) (d) (b - d)$$

Graphical Rule (Triangle Law of Addition):

If p and q are two vectors placed head-to-tail, their sum $r = p + q$ is the resultant drawn from the tail of p to the head of q .



5. SCALAR MULTIPLICATION

If k is a scalar and $a = (x, y)$, then:

$$ka \vec{a} = (kx) (ky)$$

- If $k > 1$, the vector's magnitude increases (stretches).
- If $0 < k < 1$, it shrinks.
- If $k < 0$, direction reverses.

6. MAGNITUDE AND DIRECTION OF A VECTOR

For any vector

$$\vec{a} = (x) (y)$$

- **Magnitude (Length):**

$$|\vec{a}| = \sqrt{x^2 + y^2}$$

- **Direction (Angle with x-axis):**

$$\tan \theta = y/x$$

⚠ Caution: Use the signs of x and y to determine the correct quadrant for θ .

Quadrant	Signs of (x, y)	Angle Formula
I	(+, +)	$\theta = \alpha$
II	(-, +)	$\theta = 180^\circ - \alpha$
III	(-, -)	$\theta = 180^\circ + \alpha$
IV	(+, -)	$\theta = 360^\circ - \alpha$

7. UNIT VECTORS

A unit vector has magnitude = 1 and points in a specific direction.

To find the unit vector $\hat{a}$ in the direction of vector a :

$$\hat{a} = \vec{a} / |\vec{a}|$$

Example: If $\vec{a} = 3\mathbf{i} + 4\mathbf{j}$, then $|\vec{a}| = 5$, so

$$\hat{a} = (1/5)(3\mathbf{i} + 4\mathbf{j}) = (3/5)\mathbf{i} + (4/5)\mathbf{j}$$

ROBUST WORKED EXAMPLES

Example 1: Vector Operations

Given $\mathbf{p} = 3\mathbf{i} - 4\mathbf{j}$ and $\mathbf{q} = -2\mathbf{i} + 5\mathbf{j}$.

Find: (a) $\mathbf{p}$ and $\mathbf{q}$ as column vectors. (b) $\mathbf{p} + \mathbf{q}$. (c) $3\mathbf{p} - 2\mathbf{q}$.

Solution:

a) $\mathbf{p} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$

b) $\mathbf{p} + \mathbf{q} = \begin{pmatrix} 3 \\ -4 \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \mathbf{i} + \mathbf{j}$

c) $3\mathbf{p} - 2\mathbf{q} = 3\begin{pmatrix} 3 \\ -4 \end{pmatrix} - 2\begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 9 \\ -12 \end{pmatrix} - \begin{pmatrix} -4 \\ 10 \end{pmatrix} = \begin{pmatrix} 13 \\ -22 \end{pmatrix}$

Answer: $13\mathbf{i} - 22\mathbf{j}$

Example 2: Magnitude and Direction

Find the magnitude and direction of

$$\mathbf{v} = -5\mathbf{i} + 12\mathbf{j}$$

Solution:

$$|\mathbf{v}| = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$$

$$\tan \theta = 12/(-5) = -2.4$$

Since $x < 0$, $y > 0$, vector is in 2nd Quadrant:

$$\alpha = \arctan(2.4) = 67.38^\circ, \theta = 180^\circ - 67.38^\circ = 112.62^\circ$$

Example 3: Unit Vector

Find the unit vector in the direction of $\mathbf{w} = \mathbf{i} - 2\mathbf{j}$.

Solution:

$$w = (1) (-2)$$

$$|w| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$$

$$\hat{w} = (1/\sqrt{5})(1) = (1/\sqrt{5}) (-2) (-2/\sqrt{5})$$

$$= (\sqrt{5}/5)i - (2\sqrt{5}/5)j$$

ASSESSMENT (WEEK 6)

EVALUATION (IN-CLASS)

1. State the difference between a scalar and a vector. Give two examples of each.
2. Given $a = (5)$, $b = (2)$, find $a - 3b$. $(-1) (2)$
3. Find the magnitude of $p = 5i + 12j$.
4. What is a unit vector?
5. Find the unit vector in the direction of $v = (3)$. (0)

ASSIGNMENT (TAKE-HOME)

1. Given $p = 2i + 3j$ and $q = 4i - 5j$, find a vector r such that $2p + r = 3q$.
2. Find the magnitude and direction of $a = (-1)$. (-1)
3. Find the unit vector in the direction of $b = 4i - 3j$.
4. If $v = (k)$ and $|v| = 10$, find the possible values of k . (8)
5. What is the negative of the vector $\overrightarrow{AB}$? Illustrate your answer graphically.

WEEK 8: VECTORS IN TWO DIMENSIONS II

INTRODUCTION

In the previous week, we studied the algebraic operations of vectors — how to represent them using unit vectors $\mathbf{i}$ and $\mathbf{j}$, and how to perform addition and subtraction using components.

This week, we expand into the geometric and analytical understanding of vectors — how to add vectors graphically using the triangle and parallelogram laws, how to represent position vectors, how to resolve vectors into components, and how to multiply vectors using the scalar (dot) product.

These concepts form the geometric foundation for engineering, mechanics, physics, and further mathematics applications such as force equilibrium, work done, and motion in two dimensions.

CONTENT BREAKDOWN

1. GEOMETRIC LAWS OF VECTOR ADDITION

A. Triangle Law of Vector Addition

When two vectors are added geometrically, the Triangle Law states:

"If two vectors are represented in magnitude and direction by the two sides of a triangle taken in order, then their resultant vector is represented in magnitude and direction by the third side of the triangle taken in the opposite order."

That is, if a vector $\mathbf{a}$ is followed by $\mathbf{b}$, their resultant $\mathbf{R} = \mathbf{a} + \mathbf{b}$ is the side that closes the triangle.

Mathematically:

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$

Diagram (Triangle Law):

$$A \bullet \xrightarrow{\mathbf{a}} \bullet B \xrightarrow{\mathbf{b}} \bullet C$$

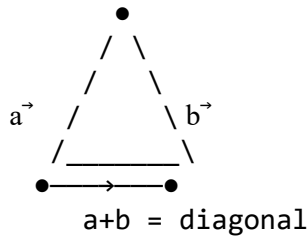
Resultant: $\overrightarrow{AC} = \mathbf{a} + \mathbf{b}$

B. Parallelogram Law of Vector Addition

If two vectors a and b are drawn from the same point (tail-to-tail), then the parallelogram law states:

"The resultant of two vectors acting at a point is represented by the diagonal of a parallelogram whose adjacent sides represent the two vectors."

Diagram (Parallelogram Law):



Thus, the resultant vector $R = a + b$ is drawn from the common tail along the diagonal.

Note: The other diagonal represents $a - b$ or $b - a$, depending on direction.

2. POSITION VECTORS AND LINE VECTORS

A position vector defines the location of a point relative to the origin.

If a point $P(x, y)$ has coordinates (x, y) , then its position vector is:

$$\vec{OP} = x\mathbf{i} + y\mathbf{j}$$

If two points $A(x_1, y_1)$ and $B(x_2, y_2)$ are given, the vector from A to B is found by:

$$\vec{AB} = \vec{OB} - \vec{OA}$$

That is:

$$\vec{AB} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j}$$

This is often summarized as:

Vector from tail to head = Position of head – Position of tail

Illustration:

The illustration shows a horizontal line with three points. The first point is labeled 0. The second point is labeled a . The third point is labeled b . A vector arrow points from 0 to a , labeled $OA = a$. Another vector arrow points from a to b , labeled $AB = b - a$.

3. RESOLUTION OF VECTORS

This process breaks a vector into horizontal (x) and vertical (y) components.

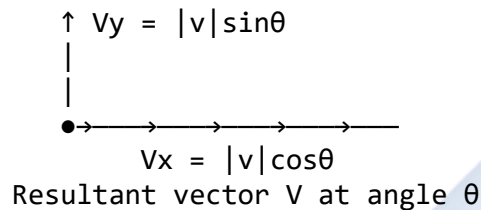
If a vector v makes an angle θ with the positive x-axis:

$$v_x = |v| \cos\theta, v_y = |v| \sin\theta$$

Thus,

$$\vec{v} = v_x \mathbf{i} + v_y \mathbf{j} = (|v| \cos\theta) \mathbf{i} + (|v| \sin\theta) \mathbf{j}$$

Illustration:



This is the foundation for analyzing forces, velocities, and accelerations in physics and engineering.

4. SCALAR (DOT) PRODUCT

The scalar product (or dot product) of two vectors is a multiplication that produces a scalar (a number).

Definition 1: Component Form

If $\vec{a} = a_1 \mathbf{i} + a_2 \mathbf{j}$ and $\vec{b} = b_1 \mathbf{i} + b_2 \mathbf{j}$, then:

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2$$

Definition 2: Geometric Form

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos\theta$$

where θ is the angle between the vectors.

5. APPLICATIONS OF THE DOT PRODUCT

A. Finding the Angle Between Two Vectors

Equating the component and geometric forms:

$$\cos\theta = (\vec{a} \cdot \vec{b}) / (|\vec{a}| |\vec{b}|)$$

Thus,

$$\theta = \cos^{-1}[(a_1 b_1 + a_2 b_2) / (\sqrt{a_1^2 + a_2^2} \sqrt{b_1^2 + b_2^2})]$$

B. Testing for Perpendicularity

If two vectors are perpendicular:

$$\theta = 90^\circ \Rightarrow \cos 90^\circ = 0$$

Hence,

$$\vec{a} \cdot \vec{b} = 0$$

This gives a simple test for perpendicularity between two vectors.

ROBUST WORKED EXAMPLES

Example 1: Position Vectors

The position vectors of points A and B are:

$$\vec{a} = 2\mathbf{i} + 5\mathbf{j}, \vec{b} = 6\mathbf{i} - \mathbf{j}$$

(a) Find $\vec{AB}$. (b) Find the position vector of the midpoint M of AB.

Solution:

(a)

$$\vec{AB} = \vec{b} - \vec{a} = (6\mathbf{i} - \mathbf{j}) - (2\mathbf{i} + 5\mathbf{j}) = 4\mathbf{i} - 6\mathbf{j}$$

(b)

$$\vec{OM} = \frac{1}{2}(\vec{a} + \vec{b}) = \frac{1}{2}((2\mathbf{i} + 5\mathbf{j}) + (6\mathbf{i} - \mathbf{j})) = 4\mathbf{i} + 2\mathbf{j}$$

Example 2: Resolution of a Vector

A force of 15 N acts at an angle of 210° to the x-axis.

Find its x and y components.

Solution:

$$F_x = 15 \cos 210^\circ = 15(-\sqrt{3}/2) = -7.5\sqrt{3}$$

$$F_y = 15 \sin 210^\circ = 15(-1/2) = -7.5$$

So,

$$\vec{F} = -7.5\sqrt{3}\mathbf{i} - 7.5\mathbf{j}$$

Example 3: Dot Product and Perpendicularity

Let $\vec{p} = 3\mathbf{i} + k\mathbf{j}$ and $\vec{q} = 2\mathbf{i} - 6\mathbf{j}$. Find k for which $\vec{p} \perp \vec{q}$.

Solution:

$$\vec{p} \cdot \vec{q} = (3)(2) + (k)(-6) = 6 - 6k$$

$$\text{For perpendicularity, } \vec{p} \cdot \vec{q} = 0$$

$$6 - 6k = 0 \Rightarrow k = 1$$

Example 4: Angle Between Two Vectors

Find the angle between $\vec{a} = 2\mathbf{i} + \mathbf{j}$ and $\vec{b} = 3\mathbf{i} - \mathbf{j}$.

Solution:

$$\vec{a} \cdot \vec{b} = 2 \times 3 + 1 \times (-1) = 5$$

$$|\vec{a}| = \sqrt{2^2 + 1^2} = \sqrt{5}, |\vec{b}| = \sqrt{3^2 + (-1)^2} = \sqrt{10}$$

$$\cos\theta = 5/(\sqrt{5}\sqrt{10}) = 1/\sqrt{2} \Rightarrow \theta = 45^\circ$$

EVALUATION (IN-CLASS QUESTIONS)

1. State and explain the Triangle Law of vector addition using a labeled diagram.
2. If $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$, express $\vec{BA}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
3. A vector has magnitude 20 and direction 120° . Find its horizontal and vertical components.
4. Compute $\vec{a} \cdot \vec{b}$ for $\vec{a} = 5\mathbf{i} - 2\mathbf{j}$ and $\vec{b} = 3\mathbf{i} + 6\mathbf{j}$.
5. What condition must two vectors satisfy to be perpendicular?

ASSIGNMENT (TAKE-HOME TASKS)

1. The position vectors of P, Q, and R are $P(1, 3)$, $Q(4, 5)$, $R(0, -2)$. Find: a) $\vec{PQ}$ b) $\vec{QR}$ c) $|\vec{PR}|$
2. Find the angle between $\vec{u} = (1, 1)$ and $\vec{v} = (1, -4)$ to the nearest degree.
3. For $\vec{a} = t\mathbf{i} - 2\mathbf{j}$ and $\vec{b} = (t+3)\mathbf{i} + 4\mathbf{j}$, find all t for which $\vec{a} \perp \vec{b}$.
4. If $|\mathbf{a}| = 4$, $|\mathbf{b}| = 5$, and the angle between them is 60° , find $\mathbf{a} \cdot \mathbf{b}$.
5. In parallelogram OABC, if $\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$: a) Express $\vec{OB}$ in terms of $\mathbf{a}$ and $\mathbf{c}$. b) Express $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.

WEEK 9–10: REVISION

INTRODUCTION

The final two weeks of this term are dedicated to consolidating your understanding of all the topics studied in the first term. By this point, you should be able to connect Trigonometry, Logic, Coordinate Geometry, and Vectors — four fundamental areas that form the core of Further Mathematics.

Revision is not just rereading; it means revisiting, reworking, and reflecting on the principles, methods, and problem-solving techniques you have learned.

REVISION STRATEGY

1. Review Core Concepts

Revisit your notes for each topic. Create a formula summary sheet like the example below:

TOPIC	KEY CONCEPTS	ESSENTIAL FORMULAS
Trigonometry	Ratios and Identities	$\sin^2\theta + \cos^2\theta = 1$, $\tan\theta = \sin\theta/\cos\theta$, $\sin(90^\circ - \theta) = \cos\theta$
	Special Angles	30° , 45° , 60° values: $\sin 30^\circ = \frac{1}{2}$, $\cos 45^\circ = \frac{\sqrt{2}}{2}$, $\tan 60^\circ = \sqrt{3}$
Logic	Connectives, Truth Tables	$\wedge$ (and), $\vee$ (or), $\rightarrow$ (implies), $\leftrightarrow$ (equivalent), $\neg$ (not)
	Basic Tables	5 core truth tables — for $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, $\neg$
Coordinate Geometry	Straight Line, Gradient	$m = (y_2 - y_1)/(x_2 - x_1)$
	Distance Formula	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
	Midpoint Formula	$M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$
	Parallel/Perpendicular	$m_1 = m_2$ (parallel), $m_1 m_2 = -1$ (perpendicular)
Vectors	Basic Operations	$\vec{b} - \vec{a}$,
	Resolution	$\vec{v} =$
	Dot Product	$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 =$

2. Rework Examples

Go back to your class notes and attempt the worked examples again without looking at the solutions. After each attempt, check your solution against the correct one to see:

- Whether you used the right formula
- Whether your sign convention was correct
- How your reasoning compares to the step-by-step method

3. Complete All Assignments

Revisit every weekly assignment and ensure each question is attempted, corrected, and understood. Where possible, create variant questions of your own to deepen mastery.

4. Practice Mixed Problems

Mix problems from different topics. For example:

- Start a problem with vector coordinates and end with finding distance using coordinate geometry.
- Combine logic and flowcharts in computer-style reasoning.
- Solve a trigonometric equation involving vector components.

This integrated practice helps you apply multiple concepts in unfamiliar contexts — a key exam skill.

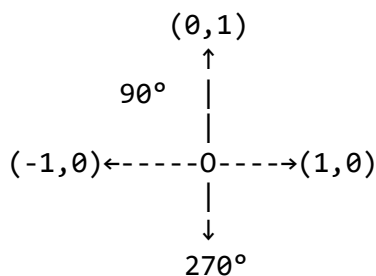
ILLUSTRATIVE REVISION SUMMARY

A. Trigonometric Review

Special Angles (Unit Circle Reference):

Angle	$\sin\theta$	$\cos\theta$	$\tan\theta$
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

Unit Circle Sketch (for visualization):



B. Logic Review

Common Connectives and Their Meanings

Symbol	Name	Meaning
$\wedge$	Conjunction	$p \wedge q$ is True only if both p and q are True
$\vee$	Disjunction	$p \vee q$ is True if at least one is True
$\rightarrow$	Implication	$p \rightarrow q$ is False only if p is True and q is False
$\leftrightarrow$	Biconditional	True if both sides have same truth value
$\neg$	Negation	Opposite of the truth value of p

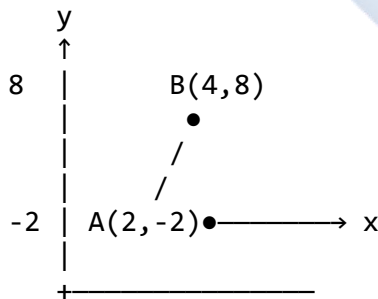
Truth Table Example (Implication and Equivalence):

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$	$(\neg p \vee q) \leftrightarrow (p \rightarrow q)$
T	T	F	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

This proves that $p \rightarrow q$ is equivalent to $\neg p \vee q$.

C. Coordinate Geometry Review

Illustration of Basic Line and Midpoint Concepts:

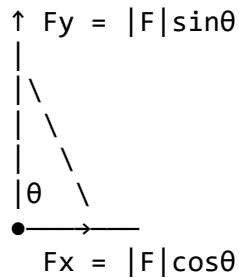


- **Midpoint (M):** $M = ((2 + 4)/2, (-2 + 8)/2) = (3, 3)$
- **Gradient (m):** $m = (8 - (-2))/(4 - 2) = 10/2 = 5$
- **Equation of line AB:** $y - 8 = 5(x - 4) \Rightarrow y = 5x - 12$

If another line is perpendicular, its gradient is $m_2 = -1/5$.

D. Vector Review

Resolution Diagram



Example:

Given $|F| = 20\text{N}$, $\theta = 30^\circ$

$$F_x = 20 \cos 30^\circ = 10\sqrt{3}, F_y = 20 \sin 30^\circ = 10$$

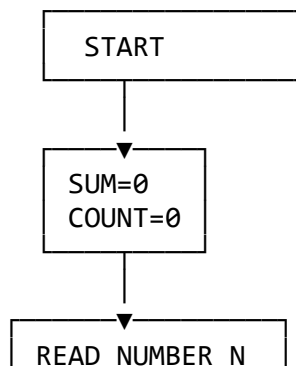
$$\therefore \vec{F} = 10\sqrt{3}\mathbf{i} + 10\mathbf{j}$$

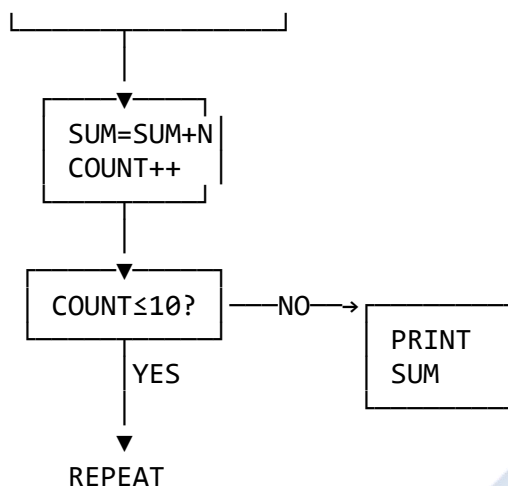
COMPREHENSIVE REVISION QUESTIONS

Part 1: Mixed Concepts

1. Evaluate: $(\tan^2 60^\circ - \sec^2 45^\circ) / (\sin 30^\circ \cos 60^\circ)$
2. Construct a truth table to show that $p \rightarrow q$ is equivalent to $\neg p \vee q$.
3. Find the equation of a line perpendicular to $x + 3y = 7$ and passing through the midpoint of $A(2, -2)$, $B(4, 8)$.
4. Given $\vec{a} = (4, -3)$ and $\vec{b} = (1, 2)$, find:
 - (a) $|2\vec{a} - \vec{b}|$
 - (b) the angle between $\vec{a}$ and $\vec{b}$.
5. Draw a flowchart to read 10 numbers and print their sum.

Flowchart (Illustration):





Part 2: Logic and Trigonometry

6. If p : True, q : False, find the truth value of $(\sim p \vee q) \leftrightarrow (p \rightarrow q)$.

7. An equilateral triangle ABC has side 8 cm. Find the exact height.

$$h = (\sqrt{3}/2) \times 8 = 4\sqrt{3} \text{ cm}$$

Part 3: Coordinate Geometry and Vectors

8. Find the area of ΔPQR with $P(0,1)$, $Q(3,4)$, $R(-1,-1)$.

$$A = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$A = \frac{1}{2}|0(4 - (-1)) + 3((-1) - 1) + (-1)(1 - 4)| = \frac{1}{2}|0 - 6 + 3| = 1.5$$

9. Find k if the line joining $(k,4)$ and $(1,2)$ is parallel to $y = 5x - 1$.

$$m = (2 - 4)/(1 - k) = 5 \Rightarrow -2/(1 - k) = 5 \Rightarrow k = 1.4$$

10. A force is resolved into 10 N East and $10\sqrt{3}$ N North. Find the magnitude and direction of the original force.

$$|F| = \sqrt{(10)^2 + (10\sqrt{3})^2} = 20 \text{ N}, \theta = \tan^{-1}(10\sqrt{3}/10) = 60^\circ$$

Thus, $|F| = 20 \text{ N}$ at 60° North of East.

ASSESSMENT (WEEK 9–10)

EVALUATION (IN-CLASS MOCK TEST)

1. Find the exact value of $\cot 45^\circ + \sec 60^\circ$.

2. Convert 53_{10} to base 2.
3. Find the gradient of a line perpendicular to $5x - 2y = 1$.
4. Find $\vec{u} \cdot \vec{v}$ where $\vec{u} = 2\mathbf{i} - 8\mathbf{j}$, $\vec{v} = 4\mathbf{i} + \mathbf{j}$.
5. Find $|\overrightarrow{PQ}|$ for $P(1, -3)$, $Q(4, 1)$.

ASSIGNMENT (TAKE-HOME FINAL REVIEW)

1. For triangle XYZ with $X(1, 4)$, $Y(3, 8)$, $Z(7, 2)$: a) Find XY . b) Find midpoint M of XZ . c) Find gradient of YZ . d) Find area of $\triangle XYZ$.
2. Given $\vec{p} = (-3, 4)$, $\vec{q} = (5, k)$: a) Find $|\vec{p}|$. b) Find the unit vector of $\vec{p}$. c) Find k if $\vec{p} \perp \vec{q}$.
3. Construct a truth table to verify if $(p \rightarrow q) \vee (q \rightarrow p)$ is a tautology.
4. Draw a flowchart that reads temperature T ($^{\circ}\text{C}$):
 - If $T < 0 \rightarrow$ Print "Freezing"
 - If $0 \leq T \leq 100 \rightarrow$ Print "Liquid"
 - If $T > 100 \rightarrow$ Print "Steam"
5. Solve for x :

$$x \tan 30^{\circ} = \sin 60^{\circ} / \cos 45^{\circ}$$

$$x = [(\sqrt{3}/2)/(\sqrt{2}/2)] / (1/\sqrt{3}) = \sqrt{6}$$

SS1 FURTHER MATHEMATICS LESSON NOTES

THIRD TERM

WEEKS 1–2: MEASURES OF LOCATION (CENTRAL TENDENCY)

INTRODUCTION

Welcome to the third term! We begin with Statistics, which is the branch of mathematics that deals with the collection, organization, analysis, and interpretation of data.

Our goal in this topic is to find the center or typical value of a set of data — where the data tends to cluster.

These are called **Measures of Location** or **Measures of Central Tendency**, and they include:

- **Mean (Arithmetic Average)** — the balance point of the data.
 - **Median (Middle Value)** — divides the data into two equal halves.
 - **Mode (Most Frequent Value)** — the value that occurs most often.
 - **Quartiles, Deciles, Percentiles** — advanced "location" measures that divide data into specific equal parts.
-

CONTENT BREAKDOWN

We will study these measures for two categories of data:

- **Ungrouped Data** – a simple list of numbers (e.g., 5, 8, 5, 7, 10).
 - **Grouped Data** – data arranged into class intervals and frequencies.
-

1. MEAN, MODE, AND MEDIAN (UNGROUPED DATA)

Mean ($\bar{x}$)

This is the arithmetic average of a data set.

$$\bar{x} = \frac{\sum x}{n}$$

Example: For data: 2, 5, 5, 8

$$\bar{x} = \frac{2 + 5 + 5 + 8}{4} = \frac{20}{4} = 5$$

Mode

The Mode is the value that occurs most frequently.

Example: Data = 2, 5, 5, 8 → Mode = 5

A dataset can be:

- **Unimodal** (one mode)
- **Bimodal** (two modes)
- **No mode** (if no value repeats)

Median

The Median is the middle value after arranging the data in ascending order.

Example (Odd number of items): Data = 2, 5, 5, 8, 10 → Median = 3rd value = 5

Example (Even number of items): Data = 2, 5, 5, 8 → Median = average of 2nd and 3rd = $(5 + 5) / 2 = 5$

Illustration: Comparing Mean, Median, and Mode

Type	Definition	Example (Data = 2, 3, 3, 6, 18)	Result
Mean	Average value	$(2+3+3+6+18)/5$	6.4
Median	Middle value	3	3
Mode	Most frequent value	3	3

2. MEAN, MODE, AND MEDIAN (GROUPED DATA)

Grouped data is presented in a frequency distribution table. We use class midpoints to compute averages.

(a) Mean for Grouped Data

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Where:

- f = frequency
- x = class midpoint
- $\sum f = N$ = total frequency

(b) Median (Q_2) for Grouped Data

To find the median, we use:

$$\text{Median} = L_1 + \frac{\left(\frac{N}{2} - cfb\right)}{f_m} \times C$$

Where:

- L_1 = lower class boundary of median class
- cfb = cumulative frequency before the median class
- f_m = frequency of the median class
- C = class size

Tip: The median class is the class interval that contains the $N/2$ th item.

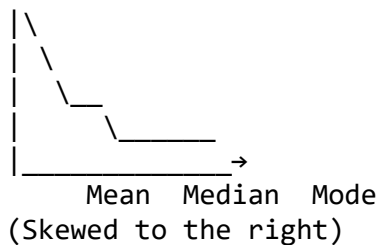
(c) Mode for Grouped Data

$$\text{Mode} = L_1 + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times C$$

Where:

- L_1 = lower boundary of modal class
- $\Delta_1 = f_m - f_{\text{before}}$
- $\Delta_2 = f_m - f_{\text{after}}$
- C = class size

Illustration: Mode, Median, and Mean of a Skewed Distribution



3. QUANTILES (QUARTILES, DECILES, PERCENTILES)

These divide the data into equal parts:

Measure	Parts	Example
Quartiles (Q)	4 parts	Q ₁ (25%), Q ₂ (50%), Q ₃ (75%)
Deciles (D)	10 parts	D ₁ (10%), D ₇ (70%)
Percentiles (P)	100 parts	P ₄₀ (40%), P ₈₀ (80%)

General Formula for Quantiles

$$P_k = L_1 + \frac{\left(\frac{kN}{100} - cfb\right)}{f_p} \times C$$

Where P_k can represent Q₁, D₇, P₄₀, etc.

Quantile	k	Position
Q ₁	25	N/4
Q ₂	50	N/2
Q ₃	75	3N/4
D ₇	70	7N/10
P ₄₀	40	2N/5

ROBUST WORKED EXAMPLE

Data Table: Marks of 80 Students

Marks (%)	10–19	20–29	30–39	40–49	50–59	60–69	70–79
Frequency (f)	4	7	10	20	25	9	5

Constructed Table

Class	f	Midpoint (x)	fx	Boundaries	Cumulative Freq (cf)
10–19	4	14.5	58.0	9.5–19.5	4
20–29	7	24.5	171.5	19.5–29.5	11
30–39	10	34.5	345.0	29.5–39.5	21
40–49	20	44.5	890.0	39.5–49.5	41
50–59	25	54.5	1362.5	49.5–59.5	66
60–69	9	64.5	580.5	59.5–69.5	75
70–79	5	74.5	372.5	69.5–79.5	80

Solutions

(a) Mean:

$$\bar{x} = \frac{3780}{80} = 47.25$$

(b) Median:

$$L_1 = 39.5, \quad N/2 = 40, \quad cfb = 21, \quad f_m = 20, \quad C = 10$$

$$\text{Median} = 39.5 + \frac{(40 - 21)}{20} \times 10 = 39.5 + 9.5 = \boxed{49.0}$$

(c) Mode:

$$L_1 = 49.5, \quad f_m = 25, \quad f_{\text{before}} = 20, \quad f_{\text{after}} = 9, \quad C = 10$$

$$\text{Mode} = 49.5 + \frac{5}{5 + 16} \times 10 = 49.5 + 2.38 = \boxed{51.88}$$

(d) Q₁:

$$L_1 = 29.5, \quad N/4 = 20, \quad cfb = 11, \quad f_q = 10$$

$$Q_1 = 29.5 + \frac{(20 - 11)}{10} \times 10 = 29.5 + 9 = \boxed{38.5}$$

(e) D₇:

$$L_1 = 49.5, \quad 7N/10 = 56, \quad cfb = 41, \quad f_d = 25$$

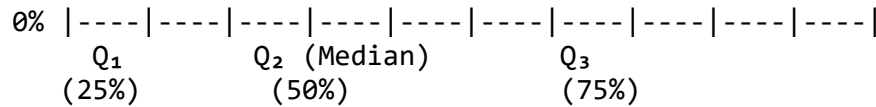
$$D_7 = 49.5 + \frac{(56 - 41)}{25} \times 10 = 49.5 + 6 = \boxed{55.5}$$

(f) P₄₀:

$$L_1 = 39.5, \quad 40N/100 = 32, \quad cfb = 21, \quad f_p = 20$$

$$P_{40} = 39.5 + \frac{(32 - 21)}{20} \times 10 = 39.5 + 5.5 = \boxed{45.0}$$

Visual Concept Summary



REVISION QUESTIONS (WEEKS 1–2)

1. What is the main difference between a measure of central tendency and a quantile?
 2. For the data set 10, 8, 15, 8, 12, find the mean, mode, and median.
 3. Why is the median often a better measure than the mean for skewed data?
 4. In a grouped frequency table, what is the modal class?
 5. State the relationship between Q_2 , D_5 , and P_{50} .
-

ASSESSMENT (WEEKS 1–2)

EVALUATION (IN-CLASS)

1. Define Mean, Median, and Mode.
 2. Find the median of: 20, 25, 21, 30, 22, 27.
 3. In the grouped median formula, what does L_1 represent?
 4. What does cfb represent?
 5. Write the formula for the position of the 3rd decile (D_3).
-

ASSIGNMENT (TAKE-HOME)

1. The ages of 10 students are: 15, 16, 15, 17, 18, 16, 15, 17, 15, 16.
 - a. Find the mean age.
 - b. Find the median age.
 - c. Find the modal age.
2. Using the "Robust Worked Example" data, calculate the Upper Quartile (Q_3).
3. The following table shows the time taken (in minutes) for 30 students to solve a puzzle.

Time (min)	0–4	5–9	10–14	15–19	20–24
Frequency (f)	3	5	10	7	5

Find:

- a. The Mean time
- b. The Median time
- c. The Modal time
- d. The 80th Percentile (P_{80})

WEEK 3–4: MEASURES OF DISPERSION

INTRODUCTION

In the last lesson, we studied measures of central tendency (mean, median, and mode), which describe the center of a data set. However, the "center" alone doesn't show how data values are spread out. Two data sets can have the same mean but be very different in variability.

Example:

Data Set	Values	Mean	Observation
A	10, 10, 10, 10, 10	10	All values are the same — no spread.
B	1, 5, 10, 15, 19	10	Values vary widely — large spread.

Conclusion: Measures of dispersion (or spread) describe how far data values are from each other or from the mean.

CONTENT

1. RANGE

Definition: The Range is the simplest measure of dispersion. It shows the difference between the largest and smallest values in a data set.

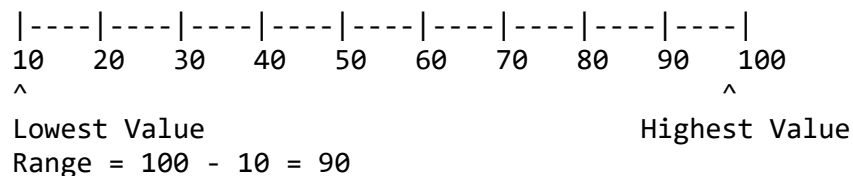
Formulas:

- **For Ungrouped Data:** $\text{Range} = \text{Highest Value} - \text{Lowest Value}$
- **For Grouped Data:** $\text{Range} = \text{Upper Boundary of Highest Class} - \text{Lower Boundary of Lowest Class}$

Weakness: The range is easily distorted by extreme values (outliers). For example, in the set [10, 11, 12, 13, 100], one unusually large value inflates the range.

Illustration:

Number Line Representation



2. INTERQUARTILE RANGE (IQR)

Definition: The Interquartile Range (IQR) measures the spread of the middle 50% of the data — it ignores the lowest 25% and highest 25%.

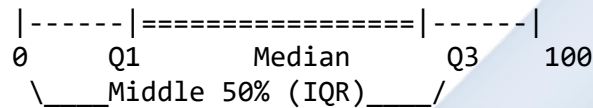
Formula: $IQR = Q_3 - Q_1$

where:

- Q_1 = Lower Quartile (25th percentile)
- Q_3 = Upper Quartile (75th percentile)

Semi-Interquartile Range (Quartile Deviation): $\text{Semi-IQR} = \frac{Q_3 - Q_1}{2}$

Box-and-Whisker Illustration:



Interpretation: A larger IQR means the middle half of the data is more spread out. A smaller IQR means the data are more consistent.

3. MEAN DEVIATION (M.D.)

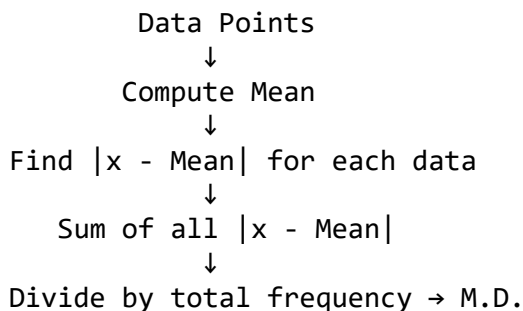
Definition: The Mean Deviation shows how far each observation deviates (differs) from the mean — on average.

Formulas:

- **Ungrouped Data:** $M.D. = \frac{\sum |x - \bar{x}|}{n}$
- **Grouped Data:** $M.D. = \frac{\sum f |x - \bar{x}|}{\sum f}$

Note: The absolute value symbol ($|x - \bar{x}|$) ensures all deviations are positive — we only measure distance, not direction.

Concept Diagram:



Interpretation:

- Small M.D. → Data points are close to the mean (low variability).
- Large M.D. → Data points are widely scattered (high variability).

4. STANDARD DEVIATION (S.D. or σ)

Definition: The Standard Deviation is the most important and widely used measure of dispersion. It represents the average distance of data points from the mean but gives more weight to large deviations because the differences are squared.

Formulas:**For Grouped Data:**

- **Method 1 (Using Mean):** $\sigma^2 = \frac{\sum f(x-\bar{x})^2}{\sum f}$
- **Method 2 (Direct Method):** $\sigma^2 = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2$

Then: $\sigma = \sqrt{\text{Variance}} = \sqrt{\sigma^2}$

Variance (σ^2) measures the average of squared deviations from the mean.

Concept Diagram:

Data → Mean → (x - mean) → Square → Sum → Divide by N → $\sqrt{\quad}$ → Standard Deviation

Interpretation:

- Small σ → Data are closely clustered around the mean.
- Large σ → Data are widely spread.

ROBUST WORKED EXAMPLE (WEEKS 3–4)

Using the data from Week 1–2 (N = 80, mean = 47.25, $Q_1 = 38.5$):

Step 1: Calculate Q_3

$$Q_3 \text{ position} = \frac{3N}{4} = 60^{\text{th}} \text{ value}$$

Q_3 class = 50–59

$L_1 = 49.5$, cfb = 41, $f(Q_3) = 25$, C = 10

$$Q_3 = 49.5 + \left(\frac{60 - 41}{25} \times 10\right) = 57.1$$

(a) Range: $79.5 - 9.5 = 70$

(b) IQR: $Q_3 - Q_1 = 57.1 - 38.5 = 18.6$

(c) Mean Deviation (M.D.):

Class	f	x	fx	x^2	fx^2	$ x - \bar{x} $	$f x - \bar{x} $
10–19	4	14.5	58	210.25	841	32.75	131.0
20–29	7	24.5	171.5	600.25	4201.75	22.75	159.25
30–39	10	34.5	345	1190.25	11902.5	12.75	127.5
40–49	20	44.5	890	1980.25	39605	2.75	55.0
50–59	25	54.5	1362.5	2970.25	74256.25	7.25	181.25
60–69	9	64.5	580.5	4160.25	37442.25	17.25	155.25
70–79	5	74.5	372.5	5550.25	27752.5	27.25	136.25
Total	80		3780		196001.25		945.5

$$\text{M.D.} = \frac{945.5}{80} = 11.82$$

(d) Standard Deviation (σ):

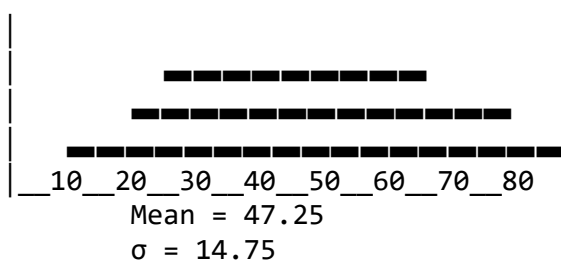
$$\begin{aligned}\sigma &= \sqrt{\frac{\sum fx^2}{N} - (\bar{x})^2} \\ &= \sqrt{\frac{196001.25}{80} - (47.25)^2} = \sqrt{217.45} = 14.75\end{aligned}$$

Interpretation:

- The average student's score deviates about 11.82 marks from the mean.
- The overall spread of scores (SD) is 14.75 marks, showing moderate variability.

Visual Summary:

Spread of Scores (Histogram Sketch)



REVISION QUESTIONS (WEEKS 3–4)

1. What is the main weakness of using the Range as a measure of dispersion?
 2. How does the Interquartile Range (IQR) solve this problem?
 3. What is the difference between Mean Deviation and Standard Deviation?
 4. Why is Variance (σ^2) always positive?
 5. If $Q_1 = 20$ and $Q_3 = 50$, find the Semi-Interquartile Range.
-

ASSESSMENT (WEEKS 3–4)

EVALUATION (IN-CLASS)

1. Define Dispersion.
 2. What is the formula for Interquartile Range?
 3. What is the formula for Mean Deviation (ungrouped data)?
 4. State the relationship between Variance and Standard Deviation.
 5. Why do we use $\sum f|x - \bar{x}|$ instead of $\sum f(x - \bar{x})$ in Mean Deviation?
-

ASSIGNMENT (TAKE-HOME)

1. For the data set 10, 12, 15, 17, 20:
 - a) Find the Range
 - b) Find the Mean ($\bar{x}$)
 - c) Find the Mean Deviation (M.D.)
 - d) Find the Standard Deviation (σ)
2. Using the data from Week 1–2 (30 students' puzzle times):
 - a) Find the Range
 - b) Find the Interquartile Range (IQR)
 - c) Find the Mean Deviation (M.D.)
 - d) Find the Standard Deviation (S.D.)

WEEK 6: MEASURES OF LOCATION — COEFFICIENT OF VARIATION (CV)

INTRODUCTION

In previous weeks, we studied Measures of Central Tendency (Mean, Median, Mode) and Measures of Dispersion (Range, Variance, Standard Deviation). This week, we combine both ideas into a single comparative tool — the Coefficient of Variation (CV).

Let's recall:

- **Mean ($\bar{x}$)** tells us the average value.
- **Standard Deviation (σ)** tells us how spread out the values are around the mean.

But if two datasets have different means, can we simply compare their standard deviations? **No!** Because one dataset might naturally have larger numbers.

For example:

Subject	Mean ($\bar{x}$)	SD (σ)
Mathematics	80	2
English	50	4

Which subject shows more consistent performance?

At first glance, Maths looks more consistent ($\sigma = 2 < 4$), but that's misleading because the means are not equal. To make a fair comparison, we need a relative measure of spread — the **Coefficient of Variation**.

CONTENT

1. DEFINITION

The **Coefficient of Variation (CV)** is a measure of relative variability that shows how large the standard deviation is compared to the mean.

It expresses the standard deviation as a percentage of the mean, making it **unitless** — which allows comparisons between data measured in different units (e.g. kilograms vs. metres).

In words:

CV measures how much variation exists in relation to the mean.

2. FORMULA

$$CV = \frac{\sigma}{\bar{x}} \times 100\%$$

Where:

- σ = Standard Deviation
- $\bar{x}$ = Mean
- CV = Coefficient of Variation (expressed as a percentage)

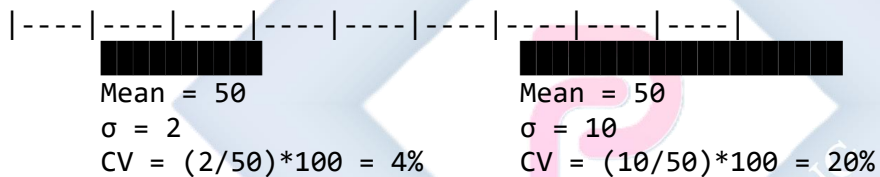
ILLUSTRATIVE DIAGRAMS

A. Graphical Comparison

Below is a visual comparison of two data sets — both centered differently but with different spreads:

Dataset A (Low CV)

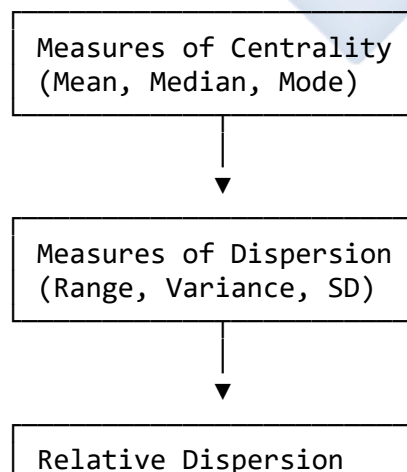
Dataset B (High CV)



Interpretation:

- Dataset A → Low CV (4%) → tightly packed data → consistent performance.
- Dataset B → High CV (20%) → widely spread data → inconsistent performance.

B. Relationship Between Measures



(Coefficient of Var.)

3. APPLICATIONS AND INTERPRETATION

The Coefficient of Variation is extremely useful in comparing the degree of variation between datasets of different means or units.

USES

- To compare consistency of students, machines, products, or experiments.
- To evaluate risk vs. return in investments.
- To check reliability or stability of measurement instruments.

INTERPRETATION

- **Low CV (0–10%)** → High consistency, low variability.
- **Moderate CV (10–30%)** → Average consistency.
- **High CV (Above 30%)** → High variability, low reliability.

ROBUST WORKED EXAMPLES

Example 1: Basic Calculation

From our previous dataset:

$$\bar{x} = 47.25, \quad \sigma = 14.75$$

Find the Coefficient of Variation.

Solution:

$$CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{14.75}{47.25} \times 100 = 31.22\%$$

Interpretation: CV = 31.22% → moderately high variability. The students' scores are fairly spread out.

Example 2: Comparing Two Subjects

Subject	Mean ($\bar{x}$)	SD (σ)
Mathematics	80	2
English	50	4

Find in which subject the student is more consistent.

Solution:

$$CV_{\text{Maths}} = \frac{2}{80} \times 100 = 2.5\%$$

$$CV_{\text{English}} = \frac{4}{50} \times 100 = 8.0\%$$

Interpretation:

- Mathematics has a lower CV → More consistent performance.
- English scores show greater variability.

Example 3: When Variance is Given

If Mean = 20, Variance = 16, find the Coefficient of Variation.

Solution:

$$\sigma = \sqrt{16} = 4$$

$$CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{4}{20} \times 100 = 20\%$$

Interpretation: The data has 20% relative variation.

CONCEPTUAL ILLUSTRATION

Imagine two car engines:

- **Engine A:** Average speed = 200 km/h, SD = 10 km/h
- **Engine B:** Average speed = 80 km/h, SD = 6 km/h

Engine	Mean Speed	SD	CV	Interpretation
A	200	10	5%	More consistent
B	80	6	7.5%	Less consistent

Even though Engine B's standard deviation (6) is smaller than A's (10), its relative variation is higher. Hence, A is the more reliable engine.

REVISION QUESTIONS

1. Define the term Coefficient of Variation.
2. Why can't we just compare standard deviations when comparing two datasets?
3. Calculate the CV for a dataset with mean 50 and SD 10.
4. Dataset A → CV = 15%, Dataset B → CV = 10%. Which dataset is more consistent?

5. What does a CV of 0% mean?
-

ASSESSMENT (WEEK 6)

EVALUATION (IN-CLASS TEST)

1. State the formula for Coefficient of Variation.
 2. If mean = 20 and variance = 16, find CV.
 3. A dataset has CV = 20% and mean = 50. Find σ .
 4. "A low CV means high consistency." (True/False).
 5. Why is the CV expressed as a percentage?
-

ASSIGNMENT (TAKE-HOME)

1. Company A: Mean salary ₱50,000, SD ₱10,000. Company B: Mean salary ₱20,000, SD ₱18,000.
 - a. Find the CV for both.
 - b. Which company has a more consistent salary structure?
 2. Brand X: Mean life : 120 hours, SD : 18 hours. Brand Y: Mean life : 100 hours, SD : 16 hours. Which brand is more reliable (more consistent)?
 3. From the dataset used in Week 3-4 (30 students' puzzle times), compute the Coefficient of Variation.
-

KEY TAKEAWAYS

- CV combines Mean and Standard Deviation into one comparative metric.
- Lower CV = Higher consistency.
- CV is unitless, so it's perfect for comparing data with different scales or units.
- Use CV in decision-making, risk analysis, and performance evaluation.

WEEK 8: HISTORY AND NATURE OF OPERATIONS RESEARCH (OR)

INTRODUCTION

This week, we move from pure statistics to **Operations Research (OR)** — a fascinating and powerful branch of Applied Mathematics often called the "Science of Better."

Operations Research is about making better decisions. It applies scientific, mathematical, and logical reasoning to complex, real-world problems where multiple factors and limitations exist.

Definition:

Operations Research (OR) is the scientific approach to decision-making that seeks to determine the best (optimal) course of action in situations involving limited resources.

In simple terms: OR helps answer questions like:

- How can we maximize profit or minimize cost?
- What is the shortest route or fastest schedule?
- How can we allocate resources most efficiently?

ILLUSTRATION: WHAT OR DOES

REAL-LIFE PROBLEM → MODEL → ANALYSIS → BEST DECISION

Example:

A company has 2 factories and 3 markets.

Goal: Minimize transport cost.

↓

Represent with equations (model)

↓

Solve mathematically

↓

Find the cheapest transport plan.

CONTENT

1. HISTORY AND MEANING

A. HISTORY OF OPERATIONS RESEARCH

Operations Research originated during **World War II (1939–1945)**. The military needed scientific ways to solve operational problems such as:

- How to locate and destroy enemy submarines efficiently.
- How to deploy radar and anti-aircraft defenses optimally.
- How to schedule convoys to minimize ship losses.

To solve these problems, scientists, mathematicians, and engineers were brought together to analyze data and propose optimal strategies. Their methods saved lives and resources.

After the war, these same techniques were adapted for business, industry, and government planning, where organizations faced similar challenges — how to make the best use of limited resources.

B. MEANING

Operations Research (OR) is the application of the scientific method to the study of complex operations and systems to make optimal decisions.

It involves:

- Building a mathematical model of a real problem.
- Using quantitative techniques to find the best solution.
- Implementing the results for maximum efficiency.

2. SCOPE (APPLICATION AREAS OF OR)

Operations Research is everywhere around us. Its tools and techniques are used in business, government, transport, manufacturing, and everyday decision-making.

Field	Example of Application	Objective
Business/Finance	Determining product mix, pricing, or stock levels	Maximize profit or minimize cost
Transportation/Logistics	Finding shortest routes for delivery vans or aircraft	Minimize travel time or fuel cost
Airlines	Scheduling flights, crew shifts, and ticket pricing	Maximize seat usage and reduce delays
Manufacturing	Deciding production schedules and resource allocation	Minimize wastage and cost
Engineering/Telecoms	Designing efficient networks or pipelines	Maximize connectivity and minimize loss

Field	Example of Application	Objective
Health Services	Assigning doctors to shifts or hospitals	Improve patient coverage
Agriculture	Allocating land and resources to crops	Maximize yield per hectare

ILLUSTRATION: OR IN ACTION

Example: Food Company (Logistics)

Problem: Deliver milk to 5 cities
Resources: 3 trucks, 1 depot
Goal: Reduce total delivery cost

↓
Build mathematical model
↓
Compute shortest and cheapest routes
↓
Result: Save 25% on transportation cost.

3. MODELS OF OPERATIONS RESEARCH

A **model** in OR is a simplified mathematical representation of a real-world situation. It helps us test and evaluate possible decisions before implementing them.

The three main OR models are:

A. LINEAR PROGRAMMING (LP)

Meaning: A mathematical technique used to find the best possible outcome — such as maximum profit or minimum cost — given certain linear constraints.

Features of LP:

- An **Objective Function** (what to maximize or minimize).
- **Constraints** (limitations on resources).
- **Decision Variables** (what you control).

Example: A bakery produces bread and cakes. Let:

- x = number of bread loaves
- y = number of cakes

Profit per bread = ₦5

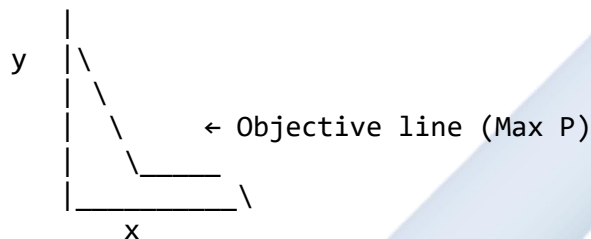
Profit per cake = ₦8

Objective function: $P = 5x + 8y$ (to maximize profit)

Constraint (based on flour use, time, etc.): $2x + 3y \leq 60$

ILLUSTRATION: LINEAR PROGRAMMING CONCEPT

Feasible Region Diagram (Graphical Method)



The shaded area (feasible region) represents all possible combinations that satisfy the constraints. The best point (corner point) gives the optimal solution.

B. TRANSPORTATION MODEL

Meaning: A model designed to determine the cheapest way to transport goods from several origins (factories) to several destinations (warehouses or depots).

Objective: Minimize total transportation cost while meeting supply and demand constraints.

Example: Dangote Cement has:

- 3 factories (A, B, C)
- 5 depots (D_1 to D_5)

Each route ($A \rightarrow D_1$, $A \rightarrow D_2$, etc.) has a cost per tonne. The Transportation Model finds the least-cost schedule to satisfy demand at each depot.

Illustration: Transportation Network

A ---]
B ---] ---> D_1, D_2, D_3, D_4, D_5
C ---]

Goal: Minimize total transport cost.

C. ASSIGNMENT MODEL

Meaning: A special form of the transportation model where the goal is to assign tasks to resources one-to-one (e.g., workers to jobs) with minimum cost or time.

Example: You have 3 workers (A, B, C) and 3 jobs (1, 2, 3). Each worker takes a different time for each job.

Worker	Job 1	Job 2	Job 3
A	3 hrs	4 hrs	2 hrs
B	2 hrs	5 hrs	3 hrs
C	4 hrs	2 hrs	3 hrs

The Assignment Model finds which worker should do which job to minimize total time.

ILLUSTRATION: ASSIGNMENT PROBLEM

WORKERS → JOBS

A → ? (2, 3, or 4 hrs?)

B → ?

C → ?

Goal: Find combination with lowest total time.

SUMMARY TABLE: COMPARISON OF OR MODELS

Model	Purpose	Key Feature	Example
Linear Programming	Optimize profit or cost	Objective function + constraints	Production planning
Transportation	Minimize distribution cost	Multiple sources and destinations	Shipping goods
Assignment	Match jobs to workers	One-to-one assignment	Task scheduling

ASSESSMENT (WEEK 8)

EVALUATION (IN-CLASS)

1. In your own words, what is Operations Research?
2. Where did Operations Research originate?
3. What is the main goal of OR? (Hint: "Opt...")
4. State two differences between a Transportation Model and an Assignment Model.

5. What two key components make up a Linear Programming problem?
-

ASSIGNMENT (TAKE-HOME)

1. **Research Task:** Find and describe one real-world application of Operations Research not mentioned in this lesson (e.g., healthcare, sports, or agriculture). Write one paragraph describing how OR is applied.
 2. **Linear Programming:** A company makes chairs and tables. Profit per chair = ₦5, profit per table = ₦8. Let x = chairs, y = tables. Write the Objective Function for profit P .
 3. **Constraint Formation:** Each chair takes 2 hours, each table takes 3 hours, and total labor available = 60 hours. Write the Constraint as a linear inequality.
 4. **Assignment Example:** Give one real-life example from your school or community where an assignment model can be applied (e.g., assigning teachers to classes, students to projects, or prefects to duties).
-

KEY TAKEAWAYS

- ✓ Operations Research = Science of better decision-making.
- ✓ Originated in World War II to optimize military operations.
- ✓ Uses mathematical models (Linear Programming, Transportation, Assignment) to solve complex problems.
- ✓ Goal: Optimization — to get the best outcome given limited resources.
- ✓ Widely applied in business, logistics, transport, health, and production planning.

WEEK 9: MODELS – LINEAR PROGRAMMING & TRANSPORTATION

INTRODUCTION

This week, we bring Operations Research (OR) to life by solving real-world optimization problems using two of its most powerful tools — **Linear Programming (LP)** and **Transportation Models**.

Operations Research helps us make the best decisions under given constraints — for example, how to:

- Maximize profit
- Minimize cost
- Utilize limited resources efficiently

Linear Programming is best for two-variable optimization problems (like production planning), while the Transportation Model is used when goods need to be distributed from multiple sources to multiple destinations at the lowest cost.

SECTION 1: LINEAR PROGRAMMING (LP)

1. Definition

Linear Programming (LP) is a mathematical technique used to find the best possible outcome (maximum profit or minimum cost) when resources are limited, and both the objective and constraints are linear.

Key Features:

- **Objective Function** → The expression to maximize or minimize.
- **Constraints** → Limitations (expressed as linear inequalities).
- **Feasible Region** → The set of all possible solutions satisfying the constraints.
- **Optimal Point** → The corner point (vertex) that gives the best value of the objective function.

2. STEPS IN SOLVING LP PROBLEMS (Graphical Method)

Step	Description	Example
1	Define the Decision Variables	Let x = no. of tables, y = no. of chairs.
2	Formulate the Objective Function	Maximize $P = 25x + 15y$

Step	Description	Example
3	Formulate the Constraints	$4x + 3y \leq 120$ (Wood) $2x + 3y \leq 72$ (Labour)
4	Include Non-negativity Constraints	$x \geq 0, y \geq 0$
5	Graph the Constraints	Convert inequalities to equalities (e.g., $4x + 3y = 120$), draw the lines, and shade infeasible regions.
6	Identify Feasible Region	The area that satisfies all constraints.
7	Find Corner Points (Vertices)	Compute where lines intersect.
8	Substitute Vertices into Objective Function	Choose the vertex with the maximum (or minimum) value.

3. WORKED EXAMPLE (Graphical Solution)

Problem:

A company makes two products – Tables (x) and Chairs (y). Profit per table = ₦25, Profit per chair = ₦15. The company is limited by:

- Wood: $4x + 3y \leq 120$
- Labour: $2x + 3y \leq 72$

Find how many tables and chairs to produce to maximize profit.

Solution Step-by-Step

Step 1: Define Variables

Let x = number of tables
y = number of chairs

Step 2: Objective Function

Maximize $P = 25x + 15y$

Step 3: Constraints

- 1 $4x + 3y \leq 120$ (Wood)
- 2 $2x + 3y \leq 72$ (Labour)
- 3 $x \geq 0, y \geq 0$

Graphical Construction

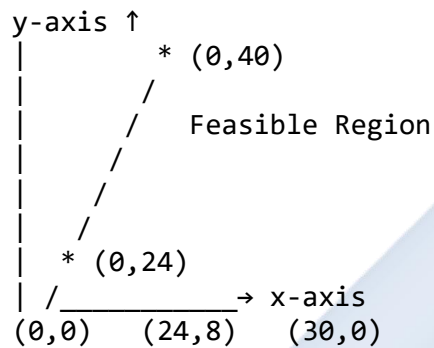
Line 1: $4x + 3y = 120$

If $x = 0 \rightarrow y = 40$
 If $y = 0 \rightarrow x = 30$
 $\rightarrow$ Points $(0,40)$ and $(30,0)$

Line 2: $2x + 3y = 72$

If $x = 0 \rightarrow y = 24$
 If $y = 0 \rightarrow x = 36$
 $\rightarrow$ Points $(0,24)$ and $(36,0)$

Illustrative Graph



Feasible Region: The shaded region below both lines and in the first quadrant.

Find Intersection Point (D)

Solve:

$$4x + 3y = 120$$

$$2x + 3y = 72$$

$$\text{Subtract: } (4x - 2x) = 120 - 72 \rightarrow 2x = 48 \rightarrow x = 24$$

$$\text{Substitute in } 2x + 3y = 72 \rightarrow 48 + 3y = 72 \rightarrow 3y = 24 \rightarrow y = 8$$

So, **D = (24, 8)**

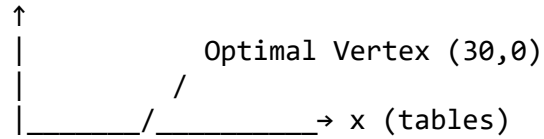
Compute Profit at Each Vertex

Vertex	Coordinates (x, y)	$P = 25x + 15y$	Profit (₦)
A	$(0,0)$	$25(0)+15(0)$	0
B	$(30,0)$	$25(30)+15(0)$	750
C	$(0,24)$	$25(0)+15(24)$	360
D	$(24,8)$	$25(24)+15(8)$	720

✓ **Maximum Profit = ₦750 at $(30, 0)$** $\rightarrow$ Produce 30 tables and 0 chairs.

Concept Diagram

Profit Plane (P)



SECTION 2: TRANSPORTATION MODELS

1. Meaning

The **Transportation Model** determines the cheapest way to distribute products from several sources (factories, plants) to several destinations (warehouses, markets), given the transportation costs between each pair.

2. Transportation Table

	W ₁	W ₂	W ₃	Supply
F ₁	5	7	8	100
F ₂	6	4	9	150
F ₃	7	8	5	120
Demand	80	110	180	370

✓ Total Supply = Total Demand = 370 → **Balanced Model**.

A. NORTHWEST CORNER (NWC) METHOD

Algorithm

- 1 Start from top-left corner (NW cell).
- 2 Allocate as much as possible.
- 3 Move right if demand is met, move down if supply is used up.
- 4 Repeat until all supply and demand are allocated.

Step-by-Step Allocation

Allocation	From	To	Allocated Units	Cost	Subtotal
1	F ₁	W ₁	80	5	400
2	F ₁	W ₂	20	7	140
3	F ₂	W ₂	90	4	360

Allocation	From	To	Allocated Units	Cost	Subtotal
4	F ₂	W ₃	60	9	540
5	F ₃	W ₃	120	5	600

✓ **Total Transportation Cost = ₦2,040**

NWC Diagram (Flow Representation)

F₁ → W₁ (80) → ₦5

└→ W₂ (20) → ₦7

F₂ → W₂ (90) → ₦4

└→ W₃ (60) → ₦9

F₃ → W₃ (120) → ₦5

B. LEAST COST METHOD (LCM)

Algorithm

- 1 Find the cell with the least cost.
- 2 Allocate as much as possible.
- 3 Cross out row/column when supply or demand is met.
- 4 Repeat until all allocations are made.

Step-by-Step Allocation

Allocation	From	To	Units	Cost	Subtotal
1	F ₂	W ₂	110	4	440
2	F ₁	W ₁	80	5	400
3	F ₃	W ₃	120	5	600
4	F ₁	W ₃	20	8	160
5	F ₂	W ₃	40	9	360

✓ **Total Cost = ₦1,960**

Note: LCM gives a cheaper solution than NWC because it considers costs.

Transportation Diagram

Factories → Warehouses (Cost per unit)

$$F_1 \rightarrow W_1 = 5, W_2 = 7, W_3 = 8$$

$$F_2 \rightarrow W_1 = 6, W_2 = 4, W_3 = 9$$

$$F_3 \rightarrow W_1 = 7, W_2 = 8, W_3 = 5$$

Optimal Allocation Flow:

$$F_2 \rightarrow W_2 = 110$$

$$F_1 \rightarrow W_1 = 80$$

$$F_3 \rightarrow W_3 = 120$$

$$F_1 \rightarrow W_3 = 20$$

$$F_2 \rightarrow W_3 = 40$$

EVALUATION (IN-CLASS QUESTIONS)

1. What is the Feasible Region in a Linear Programming problem?
2. Where does the optimal solution of a Linear Programming problem always occur?
3. What is the major difference between the Northwest Corner and Least Cost methods?
4. Why is the Least Cost Method usually more efficient?
5. In a transportation problem, what must be true about total supply and total demand?

ASSIGNMENT (TAKE-HOME TASKS)

- 1 A student must answer x questions from Section A and y questions from Section B:
 - Total questions ≤ 10
 - At least 4 from Section A
 - At least 2 from Section B
 - a. Write all inequalities.
 - b. Draw and shade the feasible region.
- 2 If questions in Section A carry 5 marks and in Section B carry 8 marks, use your graph to find the combination (x, y) that maximizes marks.
- 3 Solve this transportation problem using both NWC and LCM methods and compare the total costs:

	D ₁	D ₂	Supply
S ₁	10	12	50
S ₂	9	7	60
Demand	40	70	110

WEEK 10: APPLICATION OF MODELS (ASSIGNMENT MODEL)

TOPIC:

The Assignment Model and the Hungarian Algorithm

CONTENT

1. Introduction to the Assignment Model

The **Assignment Model** is a special type of Linear Programming Problem (LPP) used to assign a number of resources (e.g., workers, machines, or vehicles) to an equal number of tasks (e.g., jobs, locations, or duties) at the minimum total cost or maximum total efficiency.

In simple terms:

- We have n workers and n jobs.
- Each worker can perform any job, but at a certain cost (time, money, or effort).
- Our goal is to assign one worker to one job such that the total cost is minimized or total efficiency is maximized.

The Assignment Model is a special case of the Transportation Model where the number of sources = number of destinations and each has a supply and demand of one.

2. Importance and Application

Assignment models are used widely in:

- Allocating workers to jobs.
 - Assigning machines to tasks.
 - Scheduling projects and shifts.
 - Allocating delivery vehicles to routes.
 - Assigning teachers to classrooms optimally.
-

3. The Hungarian Algorithm (Method of Solution)

The **Hungarian Algorithm** provides a systematic method to find the optimal (minimum-cost) assignment. It ensures that each task is assigned to exactly one agent, and each agent handles exactly one task.

Let's go through the six main steps:

Step 1: Row Reduction

- For each row, find the smallest element.
- Subtract this smallest element from every element in that same row.
- This ensures at least one zero in every row.

This process levels all costs relative to the cheapest option for each worker.

Step 2: Column Reduction

- For each column, find the smallest element in that column.
- Subtract this smallest element from every element in that same column.
- This ensures at least one zero in every column (while keeping the row zeros).

The matrix now contains several zeros that indicate potential minimum-cost assignments.

Step 3: Test for Optimality (Cover All Zeros)

- Use the minimum number of straight lines (horizontal or vertical) to cover all zeros in the matrix.
 - If the **number of lines** = **n**, where n is the number of jobs/workers → the solution is **OPTIMAL**.
 - If **Number of Lines** < **n**, the solution is **NOT OPTIMAL**, proceed to Step 4.
-

Step 4: Improve the Matrix (if not Optimal)

- Find the smallest uncovered element (not covered by any line).
 - Subtract this value from all uncovered elements.
 - Add this value to elements covered by two lines.
 - Leave the other elements unchanged.
 - Go back to Step 3 to re-test for optimality.
-

Step 5: Repeat Until Optimal

Continue Steps 3 and 4 until the number of covering lines = n. Once this happens, the reduced cost matrix is optimal.

Step 6: Make the Assignments

- Locate a row or column with only one zero.
- Circle that zero (✓ assignment made).

- Cross out all other zeros in that row and column.
- Continue until each worker is assigned one unique job.
- Finally, compute the total cost using the original cost matrix (not the reduced one).

Worked Example

Example: A manager has 3 workers (A, B, C) and 3 jobs (J1, J2, J3). The cost (\$) for each worker on each job is:

Worker	J1	J2	J3
A	25	20	23
B	15	18	17
C	19	22	21

Find the minimum total cost assignment.

Step 1: Row Reduction

- Row A $\rightarrow \min = 20 \rightarrow [25-20, 20-20, 23-20] = [5, 0, 3]$
- Row B $\rightarrow \min = 15 \rightarrow [15-15, 18-15, 17-15] = [0, 3, 2]$
- Row C $\rightarrow \min = 19 \rightarrow [19-19, 22-19, 21-19] = [0, 3, 2]$

New Matrix:

Worker	J1	J2	J3
A	5	0	3
B	0	3	2
C	0	3	2

Step 2: Column Reduction

- J1 $\rightarrow \min = 0 \rightarrow$ no change
- J2 $\rightarrow \min = 0 \rightarrow$ no change
- J3 $\rightarrow \min = 2 \rightarrow$ subtract 2 $\rightarrow [3-2, 2-2, 2-2] = [1, 0, 0]$

New Matrix:

Worker	J1	J2	J3
A	5	0	1
B	0	3	0
C	0	3	0

Step 3: Test for Optimality

We try to cover all zeros using the minimum number of straight lines:

- Line 1: Row B (covers B–J1, B–J3)
- Line 2: Row C (covers C–J1, C–J3)
- Line 3: Column J2 (covers A–J2)

Number of lines = 3, which equals $n = 3$. ✓ **The solution is optimal.**

Step 4 & 5: Skipped (Already Optimal)

Step 6: Make Assignments

Worker	J1	J2	J3
A	5	0	1
B	0	3	0
C	0	3	0

From the table:

- A → J2 (only one zero)
- B → J1 or J3
- C → J1 or J3

There are two equivalent optimal solutions:

- A→J2, B→J1, C→J3
- A→J2, B→J3, C→J1

Both yield the same total cost.

Total Minimum Cost:

Option 1 → $20 + 15 + 21 = 56$

Option 2 → $20 + 17 + 19 = 56$

✓ **Minimum Total Cost = \$56**

Graphical Illustration

Assignment Matrix:

	JOBS		
	J1	J2	J3
A	25	20	23
B	15	18	17
C	19	22	21

After Reductions:

Row + Column zeros show potential optimal pairings

EVALUATION (CLASS ASSESSMENT)

1. Define the Assignment Model.
 2. State the major difference between the Transportation Model and the Assignment Model.
 3. Explain the purpose of the "Row Reduction" step.
 4. How do we test if the solution is optimal?
 5. What is done if the number of covering lines is less than n?
 6. Explain briefly how the final assignment is made.
-

ASSIGNMENT (HOMEWORK)

1. Solve this using the Hungarian Algorithm.

Worker	T1	T2	T3
A	10	15	9
B	9	18	5
C	6	14	3

Show all steps clearly and calculate the minimum total cost.

2. Why is column reduction important even after row reduction?
3. Mention two real-life applications of the Assignment Model.

WEEK 11: REVISION

TOPIC: GENERAL REVISION

CONTENT

This week focuses on reviewing all first term topics in Further Mathematics, with an emphasis on:

- **Statistics**
 - Mean, Median, Mode from grouped data.
 - Quartiles, Standard Deviation, and Coefficient of Variation.
- **Linear Programming**
 - Formulation of constraints and objective functions.
 - Graphical solution of LPP with two variables.
- **Transportation Problems**
 - North-West Corner (NWC) method.
 - Least Cost Method (LCM).
- **Assignment Model**
 - Solving using the Hungarian Algorithm.

REVISION EXERCISES

1. Statistics

A battery company tested 100 batteries and recorded the lifespans (in hours):

Lifespan (hours)	50–59	60–69	70–79	80–89	90–99
Frequency	8	15	40	30	7

Find:

- a. Mean
- b. Median
- c. Standard Deviation
- d. Coefficient of Variation

2. Linear Programming

A farmer wants to plant maize (x) and beans (y). He has 100 hectares and ₦240,000.

- Land constraint: $x + y \leq 100$
- Capital constraint: $2000x + 3000y \leq 240000$
- Profit: ₦400 per hectare of maize, ₦500 per hectare of beans

Required:

1. Formulate the LPP (objective function and constraints).
2. Draw the feasible region.
3. Determine the number of hectares of each crop for maximum profit.

3. Assignment Problem

Solve the following cost matrix using the Hungarian Algorithm:

Worker	P	Q	R
X	8	6	10
Y	10	12	11
Z	7	8	9

EVALUATION

1. State four steps of the Hungarian Algorithm.
2. How do you determine if an assignment solution is optimal?
3. Explain how the Assignment Model relates to the Transportation Model.
4. List three industrial uses of assignment optimization.
5. What does the Coefficient of Variation measure in statistics?

ASSIGNMENT

1. Prepare a summary sheet of all algorithms learned this term (Statistics, LP, Transportation, Assignment).
2. Attempt past exam questions on the Assignment Model.
3. Create a simple 4×4 assignment problem and solve it completely.